

Quadratic Refinements in Normal Play: Realization and the Observation Boundary

gpt-5.6-sol
noreply@openai.com

claude-fable-5
noreply@anthropic.com

a9lim
mx@a9l.im

August 2026

Abstract

Classical coin-turning games decompose into nim heaps, so their P -sets are kernels of $\mathbb{F}_2$ -linear maps; the same linear-kernel phenomenon underlies Fraenkel’s anncodes, Mock Turtles, and Mogul. We construct, to our knowledge, the first uniform normal-play family that strategically forces a genuinely *quadratic* P -set by impartial play, and we prove the matching observation boundary. For every finite-dimensional $\mathbb{F}_2$ -space, every alternating form B , every quadratic refinement Q of B , and every loaded input x , an explicit impartial local rule has root outcome P exactly when $Q(x) = 0$: its statics are refinement-blind, each transition observes at most one weight-one value of the refinement, and exactness is uniform over the full torsor of refinements of B . Conversely, any transcript-stable exact rule must observe directions spanning the input, so weight- w observations require at least $\text{wt}(x)/w$ of them; the construction attains the bound at $w = 1$, and block aggregation of the same rule attains the integral bound $\lceil \text{wt}(x)/w \rceil$ exactly at every fixed width. The Gold trace family $Q_a(x) = \text{Tr}(x x^{2^a})$ over number fields is the motivating game-built instance, with the Arf invariant as the exact second-player win-bias, and a closing construction reads the $\mathbb{Z}/4$ -valued Brown refinement off four partizan outcome classes. A basis-free Gold–Heisenberg multiplication on pairs of number values supplies the missing structural extension: its squares are Q_a and its commutators are the Gold polar form. Ten ingredients of these results are formally verified in Lean 4.

Keywords: combinatorial game theory; normal play; quadratic forms in characteristic two; Arf invariant; nim multiplication; Gold functions; extraspecial 2-groups; Brown invariant; formal verification.

MSC2020: 91A46 (primary); 11E04, 20D15, 94D10 (secondary).

1 Introduction

The linear ceiling of established coin-turning constructions is unusually rigid. Ferguson’s decomposition expresses every terminating coin-turning ruleset as a disjunctive sum of nim heaps [18]; Fraenkel’s annihilation games consequently produce linear error-correcting codes (“anncodes”) [19]. Mock Turtles and Mogul fit the same pattern, the latter recovering the binary Golay code, and Conway–Sloane lexicodes realize rich linear codes as Grundy-value-0 position sets [12]. In every case the P -set is the kernel of an $\mathbb{F}_2$ -linear map. This paper crosses that ceiling: it exhibits a normal-play rule that forces $\{Q = 0\}$ for every quadratic refinement Q of every alternating form, and it proves exactly how much refinement data any such rule must expose.

An access condition is essential to make the question nonvacuous. Any prescribed finite subset can be made the P -set of an ad hoc acyclic digraph, so a realization theorem without a uniformity

and information contract says nothing. Definition 1 fixes that contract: the rule’s position set, loading map, and terminal set are computed without consulting the refinement (**F1**); each move predicate and transition observes the refinement through at most c oracle queries at points of Hamming weight at most w_0 , with (w_0, c) independent of the dimension (**F2**); the semantics is one declared outcome convention (**F3**); and exactness must hold uniformly for every refinement of every form in the declared family (**N1**), within the stated budget (**N2**). The headline construction meets this contract at $(w_0, c) = (1, 1)$: public polar data may organize the board, but a transition sees at most one singleton value of the refinement.

1.1 Main results

The positive and negative results pair off as follows.

1. *Impartial realization* (Theorem 2, Proposition 1). The weighted-source WITT-FIFO rule is impartial and satisfies F1–F3/N1–N2 with $(w_0, c) = (1, 1)$, and for every finite-dimensional $\mathbb{F}_2$ -space, every alternating form B , every quadratic refinement Q of B , and every loaded input x , its normal-play root is P if and only if $Q(x) = 0$. A public Witt frame reduces the polar support graph to a matching; singleton-weighted source pairs transport the diagonal data; a pass-free FIFO strategy forces every potential pair of touch intervals to overlap; every core play has fixed even length; and one seat-symmetric tail move, present exactly at charge one, compiles the accumulated charge to a P/N outcome.
2. *The observation boundary* (Theorem 4, Corollaries 1 and 2). Any transcript-stable exact rule must observe refinement values at directions spanning the input, so weight- w observations require at least $\text{wt}(x)/w$ of them – hence, integrally, at least $\lceil \text{wt}(x)/w \rceil$. The construction attains the bound at $w = 1$ with observation support exactly the singleton directions of the input, and block aggregation of the same rule attains the integral bound exactly at every fixed width, closing the boundary at all widths.
3. *Fork padding* (Theorem 5). Extensional winning-fork tests – “there is a refinement-sensitive winning fork on every optimal play” – cannot certify that a realizer is non-evaluative: every finite normal-play tree can be padded, preserving all outcomes, so that every completed play enters such a fork.
4. *The Gold–Arf instance* (Theorem 6, Corollary 3). For the Gold trace forms $Q_a(x) = \text{Tr}(x x^{2^a})$ on finite nimber fields, the entire input datum is game-built: nim-multiplication is Conway’s Turning-Corners product, Frobenius is its diagonal, trace and XOR are disjunctive sums, and the diagonal weights $q_i = Q_a(e_i)$ admit a game-native Artin–Schreier source for every exponent a , with a closed trace-dual basis and a tower recursion in Conway’s bit basis. Realized by the main theorem, the Arf invariant with radical data becomes the exact second-player win-bias.
5. *Brown phases as intrinsic partizan outcomes* (Theorem 8). For a $\mathbb{Z}/4$ -valued quadratic refinement $q = \widehat{\ell} + 2Q$, a single partizan game with the two realized arenas as its left and right options has outcome class $\mathcal{N}, \mathcal{R}, \mathcal{P}, \mathcal{L}$ according to $q(x) = 0, 1, 2, 3$: the Brown residue is read off by outcome observation alone, with no external product or terminal evaluator.
6. *The game-native Gold–Heisenberg extension* (Theorem 9). The oriented half-polar term $\phi_{\lambda,a}(x, y) = \text{Tr}(\lambda x y^{2^a})$ is a bilinear cocycle. On the two-register game-value positions $\mathbb{F}_2 \times \mathbb{F}_{2^m}$ it defines one fixed noncommutative multiplication whose squares are $Q_{\lambda,a}$ and whose commutators are its polar form. For the unscaled power-of-two Gold family the full form is necessarily singular; quotienting the trace-cocycle group by the canonical polar radical gives the genuine

extraspecial core, including the Q_8 first cell. Thus the structural extension problem closes without a diagonal table or unary Q_a oracle.

7. *The obstruction landscape* (Theorems A–H, 10, 11). Coin-turning sums, commutative value-worlds (including misère quotients), frame-blind symplectic rules, affine symmetry beyond the orthogonal group, bounded B -oracle access, B -local flip rules, and E -translation dynamics all provably fail to carry a quadratic refinement; and inside the declared access class, every single-board realizer degenerates to a Q -alternating ender. These results explain why the positive construction must have the shape it has – covariant public preprocessing and a genuinely larger state space.
8. *A game-exterior no-go* (Appendix A). Altman and Lipparini ask whether Clifford algebras can be built game-theoretically [2, Problem 5.3(j)]. For ambient-coherent coefficient-faithful data on the group of short partizan games, the answer is negative: two-divisibility forces all quadratic and polar values into $\bigcap_k 4^k R$, hence zero over $\mathbb{Z}$, in characteristic 2, and over $\mathbb{Z}/4$.

1.2 Related work

The closest neighboring constructions have different input semantics. Johnson’s input-setting games let the players choose the Boolean input, whereas here x is loaded data and play must reveal its quadratic value under a fixed local rule [24]. Brandenburg’s algebraic games play directly on presented groups and rings rather than compile a quadratic predicate [6]. The invariant-games program asks whether a position-independent subtraction rule has a prescribed P -set [15]; Definition 1 is an explicit stateful analogue of that invariance question, allowing FIFO memory while fixing the loading, statics, and oracle budget. Literal move-digraph invariance is too rigid here (Theorem E), so the positive theorem uses covariant public preprocessing instead.

There is also a classical graph-theoretic ancestor of the charge mechanism. The Cohn–Lempel equality, Bouchet’s isotropic systems, and the modern interlacement-matroid formulation show how chord intersections and interlacement parity produce quadratic forms over $\mathbb{F}_2$ [10, 5, 7]. What is new here is not that overlap parity is quadratic, but that normal play can force the required overlaps strategically under a bounded-access contract.

1.3 Verification

Seven independent ingredients of the realization and the boundary are kernel-checked in Lean 4 with mathlib: the diagonal trace identities, the matching FIFO strategy, the pass-free matching/tempo compiler, the adapted-basis expansion, the transcript-span bound, the block-aggregation algebra, and the fork-padding obstruction; further modules check the algebraic cores of the Brown-selector section, the Gold–Heisenberg model, and the game-exterior appendix. These are ingredient theorems: no single Lean statement constructs the complete arena, and Section 10 records the one load-bearing step of the exactness synthesis that is not yet kernel-checked. The Lean modules, together with the exhaustive and randomized machine checks cited throughout, live in the companion repository [1].

1.4 Organization

Section 2 collects preliminaries on games, nim arithmetic, and characteristic-2 quadratic forms, and defines the Gold trace family. Section 3 proves the linear ceiling that frames the problem. Section 4 states the access contract. Section 5 gives the weighted-source WITT-FIFO construction and proves the realization theorem; Section 6 proves the observation lower bound, its attainment at

every width, and the fork-padding obstruction. Section 7 works out the Gold–Arf instance, including the all-exponent diagonal source and the win-bias reading. Section 8 packages the $\mathbb{Z}/4$ Brown refinement into a partizan outcome selector. Section 9 develops the obstruction landscape – the extraspecial dictionary, the equivariance screen, and the lettered no-go theorems – that delimits every architecture short of the positive construction, and closes the structural question those obstructions carve out with the game-native Gold–Heisenberg model. Section 10 states the formal-verification boundary. Appendix A proves the game-exterior obstruction and answers Altman–Lipparini Problem 5.3(j).

2 Preliminaries

2.1 Games and nim arithmetic

A short partizan game is built recursively as $\{L \mid R\}$, and games modulo Conway equivalence form a partially ordered abelian group under disjunctive sum [11, 4]. They do not form a ring: Conway multiplication is defined on the number subclass, not on arbitrary games. A quadratic-form story over game values therefore lives on the field-like scalar subworlds – the surreals $\mathbf{No}$ and surcomplexes $\mathbf{No}[i]$ in characteristic 0, and the nimber fields in characteristic 2 – and this paper works in the last of these.

For a normal-play disjunctive sum of impartial games, the P -condition is linear in Grundy coordinates:

$$g_1 \oplus \cdots \oplus g_n = 0.$$

Nim-addition $\oplus$ is XOR, and XOR is the disjunctive sum of impartial game values. Nim-multiplication is also game-theoretic: Conway’s Turning-Corners product [11, 27] has the mex recurrence

$$x \otimes y = \text{mex} \{ (i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \leq i < x, 0 \leq j < y \}. \quad (1)$$

The companion repository [1] implements this recurrence directly for small arguments and checks it against the fast algebraic nim product. Under $\oplus$ and $\otimes$, the ordinals below 2^{2^k} form the finite field $\mathbb{F}_{2^{2^k}}$; all machine checks cited below run inside $\mathbb{F}_{2^{128}}$ and its subfields $\mathbb{F}_{2^m}$ for $m = 1, 2, 4, \dots, 128$, not in the proper-class field On_2 .

Thus the following operations are game-realizable at the level of nimber values:

$$x \oplus y, \quad x \otimes y, \quad x \mapsto x^2 = x \otimes x, \quad \text{Tr}(x) = x + x^2 + \cdots + x^{2^{m-1}}.$$

2.2 Quadratic forms in characteristic two

In characteristic 2, a quadratic form is not determined by its polar form. For basis vectors the Clifford relations are

$$e_i^2 = q_i, \quad e_i e_j + e_j e_i = b_{ij}.$$

The polar form is alternating, so $b_{ii} = 0$, but q_i can be nonzero: the pair (q, b) , not b alone, is the faithful characteristic-2 datum. This independence is the group-theoretic heart of the paper – as Section 9 makes precise, q and b are the independent central data (squares and commutators) of an extraspecial-type extension.

For a nonsingular quadratic form over $\mathbb{F}_2$ with symplectic basis (a_k, b_k) , the Arf invariant is

$$\text{Arf}(Q) = \sum_k Q(a_k)Q(b_k) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic spaces at fixed dimension [13, 3]. Over a finite field of characteristic 2, the corresponding value lies in $F/\wp(F)$, with $\wp(y) = y^2 + y$; for finite fields this quotient is read by the field trace to $\mathbb{F}_2$.

The relevant counting theorem is classical: if Q is nonsingular on $\mathbb{F}_2^{2r}$, then

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\text{Arf}(Q)} 2^{r-1}. \quad (2)$$

For degenerate forms the standard radical-adjusted count applies: an anisotropic radical balances the values, while an isotropic radical scales the bias of the nonsingular core.

Remark 1. Ovsienko [30] identifies a precise bridge between Dickson’s classification of quadratic forms over $\mathbb{F}_2$ and the classical classification of real Clifford algebras. We use that bridge as motivation for the Arf/Clifford connection, not as a blanket classification theorem for all Clifford algebras over finite number fields. Throughout, rank and radical data are carried alongside the Arf bit, and degenerate forms are never collapsed to a single Arf value.

The small finite-field example used throughout is:

$$q = (*2, *2), \quad b_{01} = *1 \quad \text{has Arf } 1,$$

whereas

$$q = (*2, *3), \quad b_{01} = *1 \quad \text{has Arf } 0.$$

The polar entry is part of the form.

2.3 Gold trace forms

For $F = \mathbb{F}_{2^m}$ and $a \geq 1$, the Gold trace form [21] is

$$Q_a(x) = \text{Tr}(x x^{2^a}) = \text{Tr}(x^{1+2^a}).$$

Since Frobenius $x \mapsto x^{2^a}$ is additive, Q_a is a quadratic form over $\mathbb{F}_2$. Over a number field it is a composite of the game operations of Section 2.1. Throughout, “the instance (m, a) ” means Q_a on $\mathbb{F}_{2^m}$, and “ (m, a, λ) ” the scaled component $Q_{\lambda, a}(x) = \text{Tr}(\lambda x^{1+2^a})$.

Observation 1. *The Gold forms are not merely defined on a field of game values. In the finite number fields tested here, they are built from nim and game operations: Turning Corners for multiplication, diagonal Turning Corners for Frobenius squaring, and disjunctive sum for XOR and trace.*

This observation is modest but useful: it places a concrete quadratic form at the interface between impartial-game arithmetic and characteristic-2 quadratic-form theory. Section 7 develops the instance in detail – rank and Arf data, bent components, and a game-native source for the diagonal coefficients.

3 The linear ceiling

Normal-play disjunctive sums give linear P -sets, never genuine quadrics. The coin-turning architecture – the very architecture whose product theorem defines the nim product – is subject to the same ceiling.

Theorem A (coin-turning linearity). *Every Winning Ways coin-turning game has configuration P -set equal to the kernel of an $\mathbb{F}_2$ -linear nimber-valued map. In particular no coin-turning P -set is a quadric of positive polar rank.*

Proof sketch. A coin-turning position is a finite set of heads; by the Winning Ways coin-turning theory [4], the Grundy value of a configuration is the XOR of the single-coin Grundy values $g(i)$ of its heads. The P -set is therefore $\{x : \bigoplus_{i \in x} g(i) = 0\}$, the kernel of the $\mathbb{F}_2$ -linear map $x \mapsto \bigoplus_i x_i g(i)$. A machine cross-check (`octal_attack` in [1]) verifies XOR-additivity by mex recursion, exhaustively over all 32,768 companion-family choices on 4 coins. $\square$

Theorem A has a sharp classical complement: the linear ceiling is exactly where coding theory already harvests games.

Remark 2 (the lexicode shadow). Conway–Sloane lexicodes [12] are built by the greedy lexicographic rule – the mex rule – and the 1986 paper realizes them as the Grundy-value-0 positions of associated turning-game move structures; binary lexicodes are linear *because of* Sprague–Grundy theory (XOR-closure is a game theorem there, not a coding theorem). They include the Hamming codes and the length-24, $d = 8$ extended binary Golay code; over base 2^k they are closed under nim-addition, and they are linear at the Fermat-power bases 2^{2^k} – exactly the sizes at which nim-multiplication makes the ordinals below the base a field. Read together with Theorem A: lexicodes are the *floor* (rich linear codes naturally realized as Grundy-0 sets) and Theorem A is the *ceiling* (coin-turning realizes only linear P -sets); floor and ceiling meet at linear. A quadratic refinement cannot remain inside this disjunctive-sum ceiling: its obstruction is precisely the XOR-closure failure measured by the polar form. The weighted-source construction of Section 5 supplies the required quadratic realization by leaving the disjunctive-sum model.

For a quadratic form in characteristic 2,

$$Q(u + v) = Q(u) + Q(v) + B(u, v).$$

Thus for $u, v \in \{Q = 0\}$,

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The polar form is exactly the obstruction to XOR-closure. The first two layers are game-built: XOR is disjunctive sum and B is built from nim products. The play layer supplied in Section 5 integrates that bilinear data into the quadratic zero set.

4 The access contract

Since any prescribed finite subset is the P -set of some ad hoc acyclic digraph, “realizing $\{Q = 0\}$ ” only has content relative to a declared access class. Informally, three tiers bracket the problem. At one extreme, rules blind to everything but the symplectic structure cannot realize a quadric at all (Proposition 4 in Section 9); at the other, a per-position evaluation circuit for Q realizes it tautologically. The substance lies in the middle, and two natural first attempts at pinning it down fail for structural reasons. *Equivariance fails*: by Theorem E in Section 9 there is no invariance class strictly between symmetry groups that kill the P -set and symmetry groups that force extensional factoring through Q - and B -evaluation. *Vocabulary quarantine fails*: the Gold diagonal $q_i^{(m,1)} = \text{Tr}(e_i^3)$ is expressible as $\text{Tr}(\wp(w)e_i)$ using only XOR, nim-product, and trace (Proposition 2 in Section 7) – the licensed operation chain itself – so no principled line separates “computing q ” from “computing Q ” on vocabulary alone. The criterion below replaces both with access metering and a sharp transcript-observation boundary.

Definition 1 (uniform local realization). Fix the coin frame $\{e_i\}$ (the Grundy basis $g(n) = 2^n$, game-natural). *Public data*: the frame, $\oplus$, nim-product, Frobenius, and the full alternating form B (game-built via Turning Corners, Frobenius, and trace). *Quarantined data*: the quadratic refinement, presented as a weight-bounded oracle $O_q(z) = Q_q(z)$ for $\text{wt}(z) \leq w_0$. A *uniform local rule* satisfies:

- F1** (*q-blind statics*) position set, loading map, and terminal set are computed without oracle access;
- F2** (*metered access*) each move predicate and state transition together use at most c adaptive oracle queries, each at a point of Hamming weight at most w_0 , with (w_0, c) constants independent of m ;
- F3** (*declared semantics*) one canonical outcome labeling (normal, misère, loopy, or q -blind terminal payoff).

It is an *exact local realizer* if additionally:

- N1** (*torsor uniformity*) for every m , every B in the declared family, and every refinement q of B : $\text{outcome}(\iota(x)) = \text{target}$ iff $Q_q(x) = 0$;
- N2** (*locality budget*) the (w_0, c) metering of F2.

These are objective properties of a declared rule and oracle interface. They do not, by themselves, turn the informal adjective “natural” into a semantic invariant.

The next two sections give the two halves of the story. Section 5 constructs an exact local realizer with $(w_0, c) = (1, 1)$ – the smallest conceivable budget. Section 6 proves that any transcript-stable exact rule must observe refinement directions spanning the input, so the construction’s observation support is optimal, and that no constant total observation budget is possible in unbounded dimension.

5 The weighted-source Witt–FIFO construction

The polar form is public data, so it may be put into a deterministic Witt frame before play. This changes the strategic support graph from an arbitrary induced graph to a matching. A naive version would attach the dense value $Q(f_i)$ to a Witt-basis coin f_i , violating N2 in the original Grundy frame. The decisive point is to split that value into a public polar part and weight-one source *interactions*: the refinement bits weight local overlap edges rather than being accumulated as invariant terminal labels.

Fix the original frame $e_1, \dots, e_m$ and write

$$P_B(z) = \sum_{j < k} B(e_j, e_k) z_j z_k, \quad Q_q(z) = P_B(z) + \sum_j q_j z_j, \quad q_j = Q_q(e_j).$$

Choose, by a fixed Gaussian-elimination convention depending only on B , a symplectic-plus-radical basis

$$f_i = \sum_j C_{ji} e_j, \quad B(f_{a_h}, f_{b_h}) = 1$$

whose other pairings vanish. For $x = \sum_i y_i f_i$, put

$$p_i = P_B(f_i) = \sum_{j < k} C_{ji} C_{ki} B(e_j, e_k).$$

All of C , $y = C^{-1}x$, and p_i are refinement-blind public data.

For completeness, the adapted basis exists by the usual two-line induction for alternating forms. If $B = 0$, retain any basis as radical vectors. Otherwise choose the lexicographically first pair u, v with $B(u, v) = 1$. The plane $H = \langle u, v \rangle$ is nonsingular, since a vector $au + bv$ orthogonal to both u and v has $a = b = 0$; hence $V = H \oplus H^\perp$. Recurse on $H^\perp$. Choosing the first available pivot and performing row operations in the fixed original-coordinate order makes the construction deterministic. It uses only the public matrix B , works unchanged in the singular case, and terminates after one dimension drop for a radical step or two for a hyperbolic-plane step.

Rule (pass-free weighted-source WITT-FIFO). The core state is $(U, \mathcal{Q}, \text{ko}, \sigma)$: the untouched coin set, a FIFO queue of open coins, a Boolean ko bit, and the accumulated charge. The refinement-blind position universe contains the coherent core states for both values of σ together with one optionless sink. Initially $\mathcal{Q}$ is empty, $\text{ko} = \text{false}$, and $\sigma = 0$.

Load a strategic coin F_i for each $y_i = 1$, with fixed public diagonal charge p_i . Join F_{a_h} to F_{b_h} exactly when both are loaded. Every loaded strategic Witt edge has weight one. For each original coordinate $x_j = 1$, load in addition a q-blind source pair S_j^0, S_j^1 . Its potential edge has local weight $q_j = O_q(e_j)$: opening one endpoint while its mate is already queued xors q_j into the charge. Source closes have charge zero.

Every loaded coin must be touched twice, first by OPEN and then by CLOSE. OPEN(v) is legal exactly when $v \in U$; it removes v from U , appends it to $\mathcal{Q}$, and sets ko to true exactly when the old queue was empty. Thus ko protects the newly opened front while another untouched coin remains available. A subsequent OPEN clears ko because its old queue is nonempty. CLOSE is legal exactly when $\mathcal{Q}$ is nonempty and either ko is false or $U = \emptyset$; it removes the FIFO front and clears ko . There is no PASS. The exhausted-board disjunct is the only change from the earlier FIFO transition: once no untouched coin remains, the protected front has live degree zero and may drain immediately.

When a coin v opens, xor into σ the weight of its already-open mate, if any. Thus every strategic or source edge is charged exactly when its two touch intervals overlap. When a strategic front F_i closes, update

$$\sigma \oplus = p_i.$$

All other closes leave σ unchanged. Each FIFO move decreases the exact clock

$$2|U| + |\mathcal{Q}|$$

by one. If $U \neq \emptyset$, an OPEN is available; if $U = \emptyset \neq \mathcal{Q}$, a CLOSE is available even when ko is set. Thus the core cannot stall and has exactly two moves per loaded coin. At a drained state add one impartial move to the fixed sink exactly when $\sigma = 1$; a drained state with $\sigma = 0$ is already terminal. This move is available to whichever player is to move.

For a Gold form, Theorem 6 in Section 7 supplies $q_j = \text{Tr}((w_a^2 + w_a)e_j)$ for the source-pair weight. Definition 1 counts this as one weight-one oracle observation on an overlapping source OPEN. It does not claim that running the tower recursion for w_a , or evaluating its trace, is itself one primitive game move.

The sink edge reads only the accumulated state coordinate σ , not the oracle. The position universe and the terminal subset (the sink together with drained charge-zero states) are fixed

before any oracle answer is known. It is therefore a local impartial charge tail, not an inspection of x , B , or $Q(x)$.

Theorem 1 (pass-free FIFO matching theorem). *Let every component of a public graph G_0 be K_1 or K_2 . From the empty-queue root, either designated seat has one deterministic G_0 -dependent strategy which, for every edge-deleted scoring submatching, makes every core CLOSE have zero public live degree $\deg_U^{G_0}(f)$, where*

$$\deg_U^{G_0}(f) := |\{u \in U : \{f, u\} \in E(G_0)\}| \pmod{2}.$$

Consequently every public edge has overlapping touch intervals. The theorem holds for every finite board and needs no dummy.

Proof. Fix the least coin in the public order whenever an arbitrary coin is needed. The designated seat follows one rule: while $U \neq \emptyset$, it never closes; if the queue is empty it opens the least untouched coin; otherwise it opens the untouched public mate of the front when that mate exists, and opens the least untouched coin when it does not. Once $U = \emptyset$ it closes. Call an opponent checkpoint *safe* when every legal CLOSE has public live degree zero. Induct on $2|U| + |Q|$.

If the queue is empty, the designated OPEN sets ko. When untouched coins remain, this forbids CLOSE; when none remain, every live degree is already zero. If the queue has front f and its unique public mate u is untouched, OPEN u , erasing the only possible live neighbour of f . If no such mate exists, erasing any untouched coin preserves that fact. Thus the opponent always receives a safe checkpoint.

At a safe opponent checkpoint, an OPEN preserves the accumulated public close degree and returns the turn; a CLOSE has live degree zero by safety and likewise returns the turn. The new ko-set CLOSE branch causes no gap: its legality implies $U = \emptyset$, hence degree zero. Initially the queue is empty: the second seat receives a vacuously safe root, while the first seat uses the empty-queue OPEN. The designated seat itself never closes while an untouched coin remains, and after the untouched set is empty every close has degree zero. This proves the both-seat claim with one strategy depending only on G_0 .

Finally, if a public edge $\{a, b\}$ did not overlap, the endpoint opened first would close while the other endpoint was still untouched. Since G_0 is a matching, that CLOSE would have public live degree one, a contradiction. Conversely an overlapping pair is charged exactly once, when its second endpoint opens. $\square$

The pass-free transition, its exact clock, the newly legal exhausted-board CLOSE, the both-seat safe-front induction for every public submatching, and the impartial charge-tail compiler are kernel-checked in `formal/Ogdoad/ImpartialRealizer.lean` [1]. Its theorem is pointwise in the scoring submatching, while the displayed proof fixes the literal public policy and proves every CLOSE clean in the full potential matching. The weighted OPEN charges and their end-to-end composition with the adapted-basis identity remain the paper-level synthesis recorded in Section 10.

Theorem 2 (weighted-source exactness). *For every $\mathbb{F}_2$ -valued quadratic refinement Q_q on a finite-dimensional $\mathbb{F}_2$ -space and every input x , pass-free weighted-source WITT-FIFO is an impartial normal-play arena. Either designated seat can force terminal charge $\sigma = Q_q(x)$ by the same public policy. Its root is P if and only if $Q_q(x) = 0$.*

Proof. Quadratic expansion in the adapted basis gives

$$P_B(x) = \sum_i y_i P_B(f_i) + \sum_{i < k} y_i y_k B(f_i, f_k) = \sum_i y_i p_i + \sum_h y_{a_h} y_{b_h}.$$

The linear refinement part transports without any dense query:

$$\sum_i y_i \sum_j C_{ji} q_j = \sum_j x_j q_j.$$

Therefore

$$Q_q(x) = \sum_i y_i p_i + \sum_j x_j q_j + \sum_h y_{a_h} y_{b_h}. \quad (3)$$

Form instead the q-blind *potential matching* P : retain every loaded strategic Witt edge and every loaded source-pair edge, irrespective of its weight q_j . Apply the explicit safe-front strategy of Theorem 1 to P . The policy consults only this potential matching and the public FIFO state: when the front's potential mate is still untouched it opens that mate, and otherwise it opens any untouched coin. It is therefore the same strategy for every refinement q . The proof of the theorem shows more than cancellation of a total score: every close has zero live P -degree.

For one potential edge e , exactly one of the following occurs: its two touch intervals overlap, or the earlier-opened endpoint closes while the other endpoint is still untouched. The latter case would give that close live P -degree one, so the strategy excludes it. Consequently every potential pair overlaps. The rule's total OPEN charge is therefore directly

$$\sum_h y_{a_h} y_{b_h} + \sum_j x_j q_j,$$

with no refinement-dependent auxiliary graph or strategy. Every active strategic F_i closes exactly once and supplies $\sum_i y_i p_i$. Equation (3) therefore gives forced terminal charge $Q_q(x)$ for either designated seat.

If N coins are loaded, the exact core clock starts at $2N$, decreases by one on every move, and reaches zero only at drainage. Hence it is again the first player's turn after the core. If $Q_q(x) = 1$, the first player uses the public policy, forces $\sigma = 1$, and takes the unique impartial tail to win. If $Q_q(x) = 0$, the second player uses the same policy, forces $\sigma = 0$, and leaves the first player at an optionless drained state. Thus the root is respectively N or P . Notice that this is a strategy statement, not a claim that off-policy maximal plays all have charge $Q_q(x)$. The exact clock and the two normal-play strategy directions are kernel-checked in `formal/Ogdoad/ImpartialRealizer.lean`. $\square$

Proposition 1 (locality and observation support). *The rule satisfies F1–F3 and N1–N2 with $(w_0, c) = (1, 1)$. Its distinct refinement observations at input x are exactly the singleton directions $\{e_j : x_j = 1\}$; they are linearly independent and sum to x .*

Proof. The basis algorithm, $x \mapsto y$, both loaded support sets, the matching potential edges, public charges p_i , the core-state universe, the sink, and the terminal subset use only B, x , the fixed frame, and the stored state coordinates. Both values of σ occur in the position universe, so reachability may depend on q while the statics do not. Thus F1 holds. An OPEN involving a source pair needs at most the one singleton value $O_q(e_j)$, and only when its mate is already queued; every other move predicate and transition, including the tail, is refinement-blind. Even the forcing strategy is refinement-blind: it is the safe-front policy on the public potential matching P . This proves F2 and N2 with $(1, 1)$. Every move is available to either seat and the semantics is ordinary normal play, giving F3; Theorem 2 is N1 for every refinement of B .

There is one source pair exactly when $x_j = 1$, so the set of possible oracle directions is exactly $\{e_j : x_j = 1\}$. These vectors are independent and sum to x ; no proper subset spans x . $\square$

The oracle notation is accounting, not hidden information: q is a public parameter of the perfect-information game. The play-dependent source-pair repair is essential to the theorem’s *intensional* content, though not to the bare F1–F3/N1–N2 contract. A rule using only invariant source charges would make every complete history have the form

$$\sigma_q(h) = c_{B,x}(h) + \sum_{j:x_j=1} q_j,$$

so two refinements agreeing on $Q(x)$ produced action-labelled isomorphic game trees. In the present rule the coefficient of q_j is the play-dependent overlap of S_j^0, S_j^1 ; refinement data therefore enters the same local interaction law as the polar edges, not a history-independent terminal bit. This is an intensional design fact, not an outcome-only certification: Theorem 5 proves that the reachable and unavoidable fork properties do not provide such a certification.

Indeed the per-transition meter admits a deliberately extensional shortcut. Order the support of x and make a forced path which queries one singleton q_j per move while accumulating $P_B(x) + \sum_{j:x_j=1} q_j = Q_q(x)$; append a fixed parity correction and then one tail move exactly when required. This is already a formal-contract realizer at $(1, 1)$ and meets the transcript-span bound. It is a distributed evaluation circuit, not the strategic realization sought here. Thus the access contract proves the information boundary but does not, by itself, axiomatize “non-evaluative.” The pass-free construction is stronger: the same public matching policy works for the entire refinement torsor, and the queried bits enter as play-dependent overlap weights in the same interaction law as the polar edges.

The rank-zero boundary is left undecorated. The same weighted-source rule still realizes the linear refinement exactly; there simply are no strategic Witt edges to pretend otherwise.

The original isolated-dummy arbitrary-graph linking conjecture – the FIFO analogue on arbitrary support graphs – remains an interesting strict generalization, but it is not load-bearing here: the public Witt frame reduces precisely the required boards to the proved matching class, while weighted source pairs preserve fixed-frame locality. Its proof frontier and affine-response reductions belong to a companion note in preparation (draft: `writeups/linking_affine.tex` in [1]).

6 The observation boundary

The realization of Section 5 observes the refinement at exactly the singleton directions of the input. This section proves that no transcript-stable exact rule can do better, in a representation-invariant sense, and that extensional strategic-liveness tests cannot substitute for this boundary.

Theorem 3 (evaluator exclusion under N1+N2). *For any position x with $\text{wt}(x) > c w_0$, the at most c queried functionals span only weight- $\leq c w_0$ directions, so there exist refinements q, q' agreeing with every oracle answer but differing on $Q(x)$. Hence at dense positions no move predicate of a rule metered as in F2 factors through $Q(x)$ – by theorem, not stipulation.*

Definition 2 (transcript stability). Let $E(q, z)$ be the evaluation returned by refinement oracle q , let $S(q, x)$ be the finite set of observations selected at (q, x) , and let $R(q, x)$ be the rooted result. The triple (E, S, R) is *transcript-stable* if, for every q, q' and x ,

$$(E(q, z) = E(q', z) \text{ for every } z \in S(q, x)) \implies R(q, x) = R(q', x).$$

The observation set may depend adaptively on the original oracle q ; the definition says that agreement on the resulting transcript fixes the result. This is the definition formalized in `formal/Ogdoad/GoldNoEvaluator.lean`.

Definition 3 (bounded outcome certificate). Write $\text{N2cert}(C, w)$ when, for every refinement q and input x , there is a finite observation set $S(q, x)$ with $|S(q, x)| \leq C$ and $\text{wt}(z) \leq w$ for $z \in S(q, x)$ such that every same-polar refinement q' agreeing with q on $S(q, x)$ has the same rooted outcome. The set may be chosen adaptively from q ; what is bounded is the complete certificate for the outcome, not the number of queries made by one transition.

The algebra behind the next bound is standard: quadratic Boolean functions with a fixed polar form differ by linear functions, and agreement on queried linear directions is controlled by the annihilator of their span, the usual linear-structures viewpoint [9]. Linear queries are likewise the primitive of parity decision trees [40]. The novel point here is transporting that characterization to an adaptive game transcript through Definition 2.

Theorem 4 (transcript-span observation lower bound). *Let the refinement family at fixed B be closed under addition of every linear functional. If a transcript-stable rooted result is exact at x , then*

$$x \in \text{span } S(q, x).$$

In the original frame, if every observed vector has weight at most w , then

$$\text{wt}(x) \leq |S(q, x)|w.$$

Consequently no constants C, w give an exact torsor-uniform family satisfying $\text{N2cert}(C, w)$ in unbounded dimension.

Proof. If $x \notin \text{span } S$, finite-dimensional separation gives a linear functional ℓ vanishing on S but not at x . The refinements q and $q + \ell$ therefore have identical observation transcripts and rooted outcomes, while their target values at x differ, contradicting exactness. For the coordinate bound, the union of the supports of $|S|$ weight- w vectors has size at most $|S|w$ and must cover the support of x . Both forms are kernel-checked in `formal/0gdoad/GoldNoEvaluator.lean`. $\square$

Corollary 1 (optimal observation support). *At $w = 1$ the bound is attained: the weighted-source rule observes exactly the $\text{wt}(x)$ singleton directions $\{e_j : x_j = 1\}$, which span x (Proposition 1), and by Theorem 4 every transcript-stable exact weight-one interface must cover the same support.*

Block aggregation extends the attainment to every fixed width. Since $|S(q, x)|$ is an integer, Theorem 4 in fact forces $|S(q, x)| \geq \lceil \text{wt}(x)/w \rceil$; the following corollary attains that integral bound exactly, so no gap remains at any width.

Corollary 2 (block compression). *Fix $w \geq 1$. For an input x , partition $\text{supp}(x)$ into $k = \lceil \text{wt}(x)/w \rceil$ blocks of size at most w by greedy chunking in coordinate order, and write $z_1, \dots, z_k$ for the block indicator vectors. Let B' on $\mathbb{F}_2^k$ be the Gram data $B'(e'_i, e'_j) = B(z_i, z_j)$ and let Q' be the restriction of Q along $s \mapsto \sum_{i \in s} z_i$, so that Q' is a quadratic refinement of B' with diagonal $Q'(e'_i) = Q(z_i)$. The rule that plays the weighted-source WITT-FIFO arena of Theorem 2 on $(\mathbb{F}_2^k, B', Q')$ at loaded input $\mathbf{1}$, implementing each induced singleton query $Q'(e'_i)$ as the original-frame query $Q(z_i)$, satisfies F1–F3/N1–N2 at $(w_0, c) = (w, 1)$; its root outcome is P iff $Q(x) = 0$; and its possible oracle directions at input x are exactly $z_1, \dots, z_k$: pairwise disjointly supported, linearly independent, summing to x , of total weight exactly $\text{wt}(x)$, and of cardinality exactly $\lceil \text{wt}(x)/w \rceil$.*

Proof. Polarization iterates to

$$Q\left(\sum_{i \in s} z_i\right) = \sum_{i \in s} Q(z_i) + \sum_{\substack{i < j \\ i, j \in s}} B(z_i, z_j),$$

so Q' is the quadratic map on $\mathbb{F}_2^k$ with diagonal $Q(z_i)$ and polar form B' , which is alternating because B is; and $Q'(\mathbf{1}) = Q(z_1 + \dots + z_k) = Q(x)$. The induced-instance algebra – the diagonal, the Gram polar form, alternation, and the all-ones identity – together with the indicator-block bookkeeping (disjoint nonempty blocks sum to the indicator of their union and are linearly independent) is kernel-checked in `formal/Ogdoad/GoldBlockCompression.lean`.

For the contract, define one rule on the original oracle interface as the q -blind disjoint union of the induced arenas indexed by the immutable header $(x, Z_1, \dots, Z_k, B')$. The header – hence the position set, loading map, and terminal set – is computed from (B, x) and the fixed frame only, so F1 holds. Each transition of the induced arena consults the refinement through at most one singleton value $Q'(e'_i) = Q(z_i)$, one oracle query at an original-frame point of weight $|Z_i| \leq w$, giving F2 and N2 at $(w, 1)$; the semantics is the unchanged pass-free impartial tail convention, giving F3. Exactness is Theorem 2 applied to the induced instance – B' is alternating, Q' is one of its refinements, and $\mathbf{1}$ is a loaded input, so the instance lies inside the theorem’s quantified family – combined with $Q'(\mathbf{1}) = Q(x)$; this holds for every refinement Q of every B at every x , giving N1. At $x = 0$ the header has $k = 0$, the possible observation set is empty, and the optionless root is P , matching $Q(0) = 0$.

By Proposition 1 the set of possible oracle directions of the induced arena at $\mathbf{1}$ is exactly $\{e'_1, \dots, e'_k\}$, which the implementation reads at exactly $\{z_1, \dots, z_k\}$. The support, independence, total-weight, and cardinality claims are immediate from disjointness of the blocks. $\square$

Together with the integral form of Theorem 4, the observation complexity of exact transcript-stable realization under the contract is therefore exactly $\lceil \text{wt}(x)/w \rceil$ at every fixed width w ; Corollary 1 is the $w = 1$ case. Overlapping observation geometries cannot beat the partition count within the declared uniform contract, and a single width-tagged disjoint union of these rules serves every width at once; only genuinely cross-width compatibility conditions, none of which are imposed here, would pose a further question.

Theorem 5 (fork padding obstruction). *The properties “there exists a refinement-sensitive winning fork at a reachable position” and the stronger variant requiring that fork after every original completed (hence every optimal completed) play do not certify that a finite normal-play realizer is non-evaluative. Every finite normal-play tree can be padded, without changing its root outcome, so that every completed play enters an always- N two-action fork whose unique winning action swaps with one chosen refinement bit.*

Proof. Replace every terminal P -node by a one-option wrapper whose child is the always- N swapping fork. The wrapper is still P , so induction over the finite tree preserves every ancestor outcome. The new fork is forced after every original completed play. The construction and induction are kernel-checked as `win_padTerminals` in `formal/Ogdoad/GoldForkPadding.lean`. $\square$

Remark 3 (extensional liveness screens). A mixed-successor liveness screen – requiring both outcome classes among the successors of key positions – is defeated by dominated escape edges, while after dominance pruning every retained successor at a position has one outcome class. Counterfactual winning-action variants remain vulnerable to Theorem 5, even if their witness is required on every optimal play. This paper therefore uses the representation-invariant certificate lower bound of Theorem 4 for the negative boundary and states the positive rule’s syntax and access discipline directly. Strategic liveness remains useful descriptive evidence, not an axiom defining naturality.

7 The Gold–Arf instance

7.1 Rank, radical, and bent components

The polar rank of Gold forms is known [23]. In the tested regime $m = 2^k$ and $1 \leq a \leq k$, the general formula

$$\text{rank } B_{Q_a} = m - \gcd(2a, m)$$

specializes to $m - 2 \gcd(a, m)$ because $m/\gcd(a, m)$ is even. The classifier of [1] reproduces the general formula in all tested cases; representative rows are shown in Table 1.

field	a	Q	Arf	type	rank	formula
$\mathbb{F}_{2^4}$	1	$\text{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\text{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\text{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\text{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\text{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\text{Tr}(x^3)$	1	O^-	30	30

Table 1: Representative classifier output (`trace_form_arf` in [1]); the script checks all 15 cases with $m = 2, 4, 8, 16, 32$ and $1 \leq a \leq \log_2 m$.

Most of these forms have a nontrivial radical. The statement is therefore not “nondegenerate Gold form” but “positive-rank quadratic form with a nonsingular core whose Arf value is computed after symplectic reduction.”

The coefficient-1 Gold form is only one trace monomial. A standard trace description of the quadratic part of Boolean functions over $\mathbb{F}_{2^m}$ uses terms

$$\text{Tr}(c_i x^{1+2^i})$$

plus affine terms and, in even degree, a middle half-trace term. The monomials displayed here are still game-operation-built in the nimber fields: $c_i x^{1+2^i} = c_i \otimes x \otimes x^{2^i}$, followed by trace/XOR. We do not claim to package every Boolean quadratic function, and the survey cited next deliberately omits the middle term and affine bookkeeping. What it checks, exhaustively over $\mathbb{F}_{256}$, is that scaled Gold components

$$Q_{\lambda,a}(x) = \text{Tr}(\lambda x^{1+2^a})$$

already reach nondegenerate (bent) forms for many λ (`gold_family_survey` in [1]). For APN Gold exponents in that field, the observed count matches the classical $2(2^m - 1)/3$ bent-component count. This matters for the game question because a bent form has trivial radical: the loopy radical route collapses to $\{0\}$, and the frame-blind symplectic no-go of Proposition 4 (Section 9) applies without a degenerate layer to hide in. Route probes on a bent scaled component corroborate the access story at the example level: a B -plus-coordinate-frame rule reaches a quadric of the correct Arf class but not the specific Gold zero set, and the naive local spin-flip completion leaves the quadric variety (`bent_route` in [1]).

Remark 4 (a vacuous small target). At $m = 4$, $a = 1$ the zero set $\{Q_1 = 0\} \subset \mathbb{F}_{16}$ is the subfield $\mathbb{F}_4$, an $\mathbb{F}_2$ -subspace of dimension 2: the form is genuinely quadratic (polar rank 2), but its zero set is affine, so as a target for “is this P -set a genuine quadric” it is vacuous – a linear rule realizes it. The smallest honest unscaled instance is $(8, 1)$. The quadric-fitting layer of [1] (`quadric_fit`) reports exactly this distinction (genuine polar rank versus affine flat), and every probe cited here that consumes a fitted quadric is read with that guard in mind.

7.2 The diagonal framing

Once a coordinate frame is allowed, a canonical-looking quadratic refinement of B is

$$Q_{\text{frame}}(v) = \sum_{i < j} B(e_i, e_j) v_i v_j.$$

For the Gold polar forms tested here, the Gold form decomposes as

$$Q_a(v) = Q_{\text{frame}}(v) + \ell_{\text{diag}}(v), \quad \ell_{\text{diag}}(v) = \sum_i Q_a(e_i) v_i.$$

In the tested genuinely quadratic Gold cases up to $\mathbb{F}_{2^{16}}$, Q_{frame} has Arf 0 and the diagonal term changes the form to the Gold Arf-1 class.

Remark 5. This is an experimental pattern for the Gold polar forms tested here, not a theorem about every alternating form plus every coordinate frame. Random nondegenerate alternating forms can give a zero-diagonal frame quadric of Arf 1.

The diagonal $q_i = Q_a(e_i)$ is therefore the essential datum beyond the polar form. The next two results locate it precisely.

7.3 The all-exponent diagonal source

Proposition 2 (game-native diagonal source for $a = 1$). *Let $m = 2^k$ and let $w \in \mathbb{F}_{2^m}$ be the XOR of the Fermat coins 2^{2^t} for $t = 1, \dots, k-1$. Then the $a = 1$ Gold diagonal satisfies*

$$q_i^{(m,1)} = Q_1(e_i) = \text{Tr}(\wp(w) e_i), \quad \wp(w) = w \otimes w \oplus w,$$

i.e. the trace-dual element of the diagonal is $\wp(w)$. Verified at $m = 4, 8, 16, 32$ via the trace-pairing uniqueness of the dual element (`gold_diag_probe` in [1]).

This gives a game-native, tower-uniform, commutative source for the $a = 1$ diagonal: one nim-squaring, one XOR, one trace – all inside the licensed operation chain of Section 2.1. The following theorem closes the same sourcing question for every exponent, including even a , and identifies the exact obstruction for a general scale.

For $c \in K_m = \mathbb{F}_{2^m}$, let $\lambda_{a,c}^{(m)}$ be the unique trace-dual of the scaled canonical-basis diagonal:

$$\text{Tr}_m(\lambda_{a,c}^{(m)} e_i) = \text{Tr}_m(c e_i \text{Fr}^a(e_i)) \quad (0 \leq i < m). \quad (4)$$

Theorem 6 (sharp diagonal-source criterion). *For every admitted power-of-two degree m and every $a \geq 0$,*

$$\text{Tr}_m(\lambda_{a,c}^{(m)}) = \text{Tr}_m(c), \quad \lambda_{a,c}^{(m)} \in \wp(K_m) \iff \text{Tr}_m(c) = 0.$$

In particular, if $m = 2^k \geq 2$ and $c = 1$, there is a $w_a^{(m)} \in K_m$ such that

$$\lambda_{a,1}^{(m)} = (w_a^{(m)})^2 + w_a^{(m)}, \quad Q_a(e_i) = \text{Tr}_m(((w_a^{(m)})^2 + w_a^{(m)}) e_i)$$

for every canonical basis coin e_i .

Proof. Put $e_0 = 1$ in the defining duality (4):

$$\text{Tr}_m(\lambda_{a,c}^{(m)}) = Q_{a,c}(1) = \text{Tr}_m(c).$$

The Artin–Schreier image of a finite characteristic-two field is exactly the absolute-trace kernel [28, Thm. 2.25]. This proves the equivalence. For the unscaled form, $\text{Tr}_m(1) = m \bmod 2 = 0$ whenever the power-of-two degree satisfies $m \geq 2$. $\square$

The one-line proof is basis-independent: it works in any basis containing 1. The stronger descent and recursion below are special to Conway's bit basis. Write $m = 2M$, $K_m = K_M(u) = K_M \oplus uK_M$, where $u = e_M$. Conway's Fermat-power rules give

$$\sigma(u) = u + 1, \quad ue_j = e_{M+j}, \quad \sigma = \text{Fr}^M$$

[11, Ch. 6]. If $e_j^{*,(M)}$ denotes the trace-dual of e_j in K_M , then the trace-dual bit basis upstairs has the closed form

$$e_{M+j}^{*,(2M)} = e_j^{*,(M)}, \quad e_j^{*,(2M)} = (1 + u)e_j^{*,(M)}. \quad (5)$$

Indeed, for $A, B, e \in K_M$ the relative trace $T = 1 + \sigma$ satisfies

$$T((A + uB)e) = Be, \quad T((A + uB)ue) = (A + B)e. \quad (6)$$

Substitution of $A = 0, B = e_j^{*,(M)}$ and then $A = B = e_j^{*,(M)}$ gives the two dualities in Equation (5), including the vanishing cross blocks.

Two consequences now take one line each. Let

$$d_0 = T(c), \quad d_1 = T(cu \text{Fr}^a(u)), \quad L_s = \lambda_{a,d_s}^{(M)}.$$

The lower and upper diagonal blocks have duals L_0, L_1 , so

$$\lambda_{a,c}^{(2M)} = (1 + u)L_0 + L_1 = (L_0 + L_1) + uL_0. \quad (7)$$

For $c = 1$, the lower block is zero because its entries lie in K_M and have zero trace upstairs; hence $L_0 = 0$ and

$$\lambda_{a,1}^{(2M)} = L_1 \in K_M. \quad (8)$$

Thus the closed dual basis simultaneously explains the scaled recursion and the unscaled descent. Equations (6) and the finite-field Artin–Schreier exact sequence are kernel-checked in `formal/Ogdoad/GoldDiagonal.lean`.

The source itself may also be constructed tower-recursively, with no final linear-algebra fallback. Write the quadratic extension as

$$K_{2M} = K_M(u), \quad u^2 + u = \delta, \quad \text{Tr}_M(\delta) = 1.$$

The last equality is the Artin–Schreier irreducibility criterion. Given a trace-zero target $\mu = A + uB$, seek $w = X + uY$. Expanding gives

$$w^2 + w = (X^2 + X + \delta Y^2) + u(Y^2 + Y). \quad (9)$$

Because $\text{Tr}_M(B) = \text{Tr}_{2M}(\mu) = 0$, first solve $Y^2 + Y = B$ downstairs. The two choices are Y and $Y + 1$; replacing one by the other toggles the trace of $A + \delta Y^2$ by $\text{Tr}_M(\delta) = 1$. Choose the one for which that trace vanishes, and solve

$$X^2 + X = A + \delta Y^2$$

downstairs. Recursing to K_1 constructs w throughout the Conway tower. The two final sources differ by 1; normalize by requiring the e_0 coordinate to vanish. At $a = 1$ this normalized source is exactly the Fermat-coin source of Proposition 2, giving

$$m = 8 : 20, \quad m = 16 : 276, \quad m = 32 : 65812.$$

The uniqueness of the normalized source makes this identification immediate from Proposition 2; the displayed tower recursion makes it constructive for every exponent. The case $a = 0$ is included algebraically, but $Q_0(x) = \text{Tr}(x^2)$ is linear and has zero polar form, so it is not a genuinely quadratic Gold instance [21].

Remark 6 (subfield vanishing). $q_i^{(m,a)} = 0$ for all $i < m/2$: the Fermat basis elements at lower indices, their Frobenius iterates, and their nim products stay inside the half-field, whose absolute trace from an even-degree extension vanishes. The Gold diagonal lives on the top Fermat layer only.

Remark 7 (the framing is game-built). The diagonal framing

$$q_i = Q_a(e_i) = \text{Tr}(e_i^{1+2^a})$$

is game-built for every a : Proposition 2 gives the explicit $a = 1$ source, and Theorem 6 gives a constructive source for every exponent. The weighted-source Witt-FIFO theorem of Section 5 then supplies a local normal-play rule with P -set $\{x : Q_a(x) = 0\}$.

7.4 Arf as win-bias

Corollary 3 (Arf as win-bias). *Load weighted-source WITT-FIFO with a Gold instance (m, a) . Its P -loadings are exactly $\{x \in \mathbb{F}_{2^m} : Q_a(x) = 0\}$. On a nonsingular core the Arf invariant is the sign of the bias between second-player wins and first-player wins; in the singular case the radical data supplies the standard adjustment to the count (2).*

Proof. Immediate from Theorem 2 applied to Q_a , together with the zero-count formula (2) and its radical-adjusted degenerate form. $\square$

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Representative zero-counts (`arf_win_bias` in [1]). The bias is $\#\{Q = 0\} - 2^{m-1}$.

8 Brown phases as intrinsic partizan outcomes

The realized arenas can be packaged pairwise to read off $\mathbb{Z}/4$ -valued refinements as well. Let V be a finite $\mathbb{F}_2$ -space and let

$$q(x + y) = q(x) + q(y) + 2b(x, y), \quad q: V \longrightarrow \mathbb{Z}/4,$$

where b is symmetric bilinear. Write

$$G(q) = \sum_{x \in V} i^{q(x)}.$$

Whenever $G(q) \neq 0$, define $\beta(q) \in \mathbb{Z}/8$ by $G(q)/|G(q)| = \exp(\pi i \beta(q)/4)$. This definition includes singular b when its Gauss sum is nonzero; when the sum vanishes there is no phase, and the radical is retained as separate data.

The next result is a convenient coordinate form of standard Brown-form material: Brown's relation $2Q(u) = jB(u, u)$, Wood's classification, and the same two-digit decomposition in the generalized-bent literature [8, 39, 33]. The correlated Walsh formula is the $q = 4$ case of the 2-adic generalized-bent decomposition [35, 34].

Theorem 7 (canonical Brown split). *There are unique maps $\ell: V \rightarrow \mathbb{F}_2$ and $Q: V \rightarrow \mathbb{F}_2$ such that*

$$q = \widehat{\ell} + 2Q, \quad B_Q = b + \ell \otimes \ell,$$

where $\widehat{0} = 0$ and $\widehat{1} = 1$ in $\mathbb{Z}/4$. Here ℓ is linear and Q is an ordinary characteristic-2 quadratic form with alternating polar form B_Q . Conversely, these data recover q and b . Moreover, with $W(R) = \sum_x (-1)^{R(x)}$,

$$G(q) = \frac{1+i}{2}W(Q) + \frac{1-i}{2}W(Q + \ell). \quad (10)$$

Proof. Reduction modulo 2 makes $\ell = q \bmod 2$ linear, and the two binary digits of $q(x)$ define $Q(x)$. Comparing the two expressions for $q(x+y)$, using $\widehat{a+c} = \widehat{a} + \widehat{c} + 2ac$, gives $B_Q = b + \ell \otimes \ell$. Finally, $i^\ell = (1+i)/2 + (1-i)(-1)^\ell/2$, which gives Equation (10). $\square$

Pointwise, the two coordinates and the Brown weight are

$q(x)$	0	1	2	3
$\ell(x)$	0	1	0	1
$Q(x)$	0	0	1	1
$i^{q(x)}$	1	i	-1	$-i$

Thus Equation (10) is a correlated, rather than factorized, pair of quadratic censuses. In particular, the Brown phase is not recovered from a marginal count for ℓ and an independent Arf count for Q .

Remark 8 (singular boundary). Let $D = \text{rad } b$. The diagonal identity $b(r, r) = q(r) \bmod 2$ gives $\ell|_D = 0$ and $q|_D = 2Q|_D$. If some $r \in D$ has $Q(r) = 1$, translation by r reverses every summand of $G(q)$, so $G(q) = 0$. If $Q|_D = 0$, the pair (ℓ, Q) descends to V/D , and the definition of β above is exactly the phase on that nonsingular quotient. This is why a zero Gauss sum is not assigned an artificial Brown octant.

Proposition 3 (Brown phase in quadratic coordinates). *Suppose that b is nondegenerate. This is a re-coordinatization of the classification formulas of Wood [39].*

1. *If $\dim V$ is even, then B_Q is nondegenerate. If $\ell(x) = B_Q(a, x)$, then*

$$\beta(q) = 4 \text{Arf}(Q) + 2Q(a) \pmod{8}.$$

2. *If $\dim V$ is odd, then $\text{rad } B_Q = \langle w \rangle$, $\ell(w) = 1$, and B_Q is nondegenerate on $H = \ker \ell$. Then*

$$\beta(q) = 4 \text{Arf}(Q|_H) + \begin{cases} 1, & Q(w) = 0, \\ 7, & Q(w) = 1 \end{cases} \pmod{8}.$$

Proof. In even dimension, translation by a gives $W(Q + \ell) = (-1)^{Q(a)}W(Q)$, while $W(Q) = (-1)^{\text{Arf}(Q)2^{\dim V/2}}$; substitute these in Equation (10). In odd dimension, sum over the pairs $h, h+w$ for $h \in H$. The Gauss sum factors as

$$(-1)^{\text{Arf}(Q|_H)2^{(\dim V-1)/2}}(1 + i(-1)^{Q(w)}),$$

whose two possible final factors have octants 1 and 7. $\square$

For every ordinary quadratic form R on V , let $\mathcal{A}_R(x)$ be the finite normal-play arena of Theorem 2. The only interface used here is

$$o(\mathcal{A}_R(x)) \in \{\mathcal{P}, \mathcal{N}\}, \quad o(\mathcal{A}_R(x)) = \mathcal{P} \iff R(x) = 0.$$

In particular, no impartiality assertion about this arena is needed.

Theorem 8 (intrinsic Brown selector). *For $q = \widehat{\ell} + 2Q$, the single partizan game*

$$\mathcal{B}_q(x) = \{\mathcal{A}_{Q+\ell}(x) \mid \mathcal{A}_Q(x)\}$$

has the outcome table

$q(x)$	$\ell(x)$	$Q(x)$	$Q(x) + \ell(x)$	$o(\mathcal{A}_{Q+\ell}(x))$	$o(\mathcal{A}_Q(x))$	$o(\mathcal{B}_q(x))$
0	0	0	0	$\mathcal{P}$	$\mathcal{P}$	$\mathcal{N}$
1	1	0	1	$\mathcal{N}$	$\mathcal{P}$	$\mathcal{R}$
2	0	1	1	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{P}$
3	1	1	0	$\mathcal{P}$	$\mathcal{N}$	$\mathcal{L}$.

Thus the fixed decoder $\mathcal{N}, \mathcal{R}, \mathcal{P}, \mathcal{L} \mapsto 0, 1, 2, 3$ recovers $q(x)$.

Proof. If Left starts, her sole move enters $\mathcal{A}_{Q+\ell}(x)$ with Right to move, so Left wins exactly when $Q(x) + \ell(x) = 0$. If Right starts, his sole move enters $\mathcal{A}_Q(x)$ with Left to move, so Right wins exactly when $Q(x) = 0$. These two starter bits give the displayed four rows. $\square$

The four outcome classes are, by definition, the two truth values ($G \geq 0, G \leq 0$): $\mathcal{P} = (1, 1)$, $\mathcal{L} = (1, 0)$, $\mathcal{R} = (0, 1)$, and $\mathcal{N} = (0, 0)$. Hence the selector packages the correlated pair $(Q + \ell, Q)$ into the two starter faces of one game tree; it is not an additional algebraic invariant. The decoder is only a label bijection, and outcomes do not compose under disjunctive sum. Indeed, if the followers are replaced by the simplest games with their stated outcomes, then $\mathcal{B}_q(x)$ is respectively

$$*, \quad \downarrow, \quad 0, \quad \uparrow \quad (q(x) = 0, 1, 2, 3),$$

and $\uparrow + \downarrow = 0 \neq *$ displays the failure of additivity. The construction does not synchronize two plays: after the root move, exactly one follower is entered and ordinary normal play determines the winner.

Corollary 4 (outcome census). *If $n_{\mathcal{C}}$ counts the inputs with selector outcome $\mathcal{C}$, then*

$$G(q) = n_{\mathcal{N}} + in_{\mathcal{R}} - n_{\mathcal{P}} - in_{\mathcal{L}}.$$

Thus the four-outcome census retains exactly the correlated Walsh data and, when defined, the Brown phase.

Proof. Group the terms $i^{q(x)}$ by the table in Theorem 8. $\square$

Remark 9 (the global boundary). This is a finite-space construction. Ambient-coherent Brown data on all short-game values vanish by two-divisibility; the stronger game-exterior statement and its exact coherence hypothesis are recorded in Appendix A. The section/framing issue for a bare $\mathbb{Z}/4$ central extension is separate; its algebraic core is kernel-checked in `formal/0gdoad/BrownGame.lean`.

9 The middle tier: obstructions and the structural model

The positive theorem’s shape – covariant public preprocessing, a genuinely larger FIFO state space, play-dependent source interactions – is forced. This section delimits everything short of it: commutative provenance, frame-blind and affine symmetry, bounded B -oracle access, B -local flip rules, E -translation dynamics, and finally, inside the declared access class itself, a rigidity theorem showing that every single-board realizer degenerates. Theorem A supplied the coin-turning case. Several results are corroborated by exhaustive machine checks at small dimension, recorded with each statement; example-level probes (an interactive-kernel solver over finite move graphs, loopy Draw-set searches, bent-component route probes) narrowed the target during this work and are available in [1], but the theorems below do not depend on them.

9.1 A frame-blind no-go

There is a clean obstruction for rules that see only a nondegenerate symplectic form and no frame.

Proposition 4. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^r$ with $r \geq 2$. If a finite normal-play game has position set V and a move relation invariant under $\mathrm{Sp}(B)$, then its P -set is not a nondegenerate quadric.*

Proof. Each element of $\mathrm{Sp}(B)$ is an automorphism of the move graph, so the retrograde outcome function is $\mathrm{Sp}(B)$ -invariant. The P -set is therefore a union of symplectic orbits. For $r \geq 2$, the symplectic group is transitive on $V \setminus \{0\}$ [36], so the invariant subsets are only $\emptyset$, $\{0\}$, $V \setminus \{0\}$, and V . A nondegenerate quadric in dimension at least 4 has $2^{2r-1} \pm 2^{r-1}$ points, which is none of these. $\square$

The hypothesis matters. Coordinate-dependent rules have already broken $\mathrm{Sp}(B)$ symmetry, so this proposition does not explain every negative probe. It explains only genuinely frame-blind B -only rules. The remainder of this section extends the obstruction far beyond this symmetric class.

9.2 The extraspecial dictionary

The diagonal-framing question of Section 7.2 can be restated in group-theoretic terms, and the restatement sharpens it. The dictionary is classical – the characteristic-2 Heisenberg/extraspecial picture of quadratic forms [32, 22] – and we prove the pieces we use, to keep the paper self-contained; the lemmas of this subsection and Proposition 5 are standard mathematics, while reading E -equivariance as a naturality criterion is interpretation. The weighted-source theorem realizes the quadric without claiming that ordinary play itself is a model of the central extension E ; that stricter extension-model question remains optional and is recorded as an open question below.

Throughout, $V = \mathbb{F}_2^n$ written additively, and $Z = \{1, z\} \cong \mathbb{Z}/2$ written multiplicatively. A quadratic map $Q: V \rightarrow \mathbb{F}_2$ is a function with $Q(0) = 0$ whose polarization $B(x, y) = Q(x + y) + Q(x) + Q(y)$ is bilinear; B is then alternating (hence symmetric), and Q may have any diagonal $q_i = Q(e_i)$, exactly the char-2 datum of Section 2.2.

Lemma 1 (extraspecial dictionary). *Let $1 \rightarrow Z \rightarrow E \xrightarrow{\pi} V \rightarrow 1$ be a central extension of the elementary abelian 2-group V by $Z \cong \mathbb{Z}/2$.*

- (i) *For $x \in V$ and any lift $\tilde{x} \in \pi^{-1}(x)$ one has $\tilde{x}^2 \in Z$, and $\tilde{x}^2$ is independent of the lift; the squaring form $Q_E: V \rightarrow \mathbb{F}_2$ defined by $\tilde{x}^2 = z^{Q_E(x)}$ is well defined, as is the commutator*

pairing B_E defined by $[\tilde{x}, \tilde{y}] = z^{B_E(x,y)}$, which is bilinear and alternating. They satisfy

$$(\tilde{x}\tilde{y})^2 = \tilde{x}^2 \tilde{y}^2 [\tilde{x}, \tilde{y}], \quad \text{equivalently} \quad Q_E(x+y) = Q_E(x) + Q_E(y) + B_E(x,y),$$

so Q_E is a quadratic map with polar form B_E .

(ii) Conversely, every quadratic map Q arises: fixing a basis $e_1, \dots, e_n$ of V , the bilinear 2-cocycle

$$\varphi(x, y) = \sum_i x_i y_i Q(e_i) + \sum_{i>j} x_i y_j B(e_i, e_j)$$

defines a group $E_Q = Z \times V$ with $(s, x)(t, y) = (s + t + \varphi(x, y), x + y)$, whose squaring form is Q and whose commutator pairing is the polar form B .

(iii) The assignment $Q \mapsto [E_Q]$ is a bijection from quadratic maps on V to equivalence classes of central extensions of V by $\mathbb{Z}/2$.

(iv) E_Q is abelian iff $B = 0$ (iff Q is linear); and for $n \geq 1$, E_Q is extraspecial (i.e. $Z(E) = [E, E] = \Phi(E) \cong \mathbb{Z}/2$) iff B is nondegenerate, which forces $n = 2r$ even and $|E_Q| = 2^{1+2r}$.

Proof. (i) $\pi(\tilde{x}^2) = 2x = 0$, so $\tilde{x}^2 \in Z$; replacing $\tilde{x}$ by $z\tilde{x}$ multiplies the square by $z^2 = 1$. Commutators of lifts lie in Z because V is abelian, are central, and are unchanged by central retagging of the lifts; centrality of commutators gives bimultiplicativity, and $[\tilde{x}, \tilde{x}] = 1$ gives $B_E(x, x) = 0$. For the identity: with $c = [\tilde{y}, \tilde{x}]$ central, $\tilde{x}\tilde{y}\tilde{x}\tilde{y} = \tilde{x}(\tilde{x}\tilde{y}c)\tilde{y} = \tilde{x}^2\tilde{y}^2c$, and $c = c^{-1} = [\tilde{x}, \tilde{y}]$ since Z has exponent 2. As $\tilde{x}\tilde{y}$ lifts $x + y$, the additive identity follows.

(ii) A bilinear map is a 2-cocycle, so E_Q is a group with center containing $Z = \{(0, 0), (1, 0)\}$. Squares: $(s, x)^2 = (\varphi(x, x), 0)$ and $\varphi(x, x) = \sum_i x_i Q(e_i) + \sum_{i>j} x_i x_j B(e_i, e_j) = Q(x)$ by the polarization expansion of Q in the basis. Commutators: $(s, x)(t, y)(s, x)^{-1}(t, y)^{-1} = (\varphi(x, y) + \varphi(y, x), 0) = (B(x, y), 0)$, using that the symmetrization of φ is B (the diagonal terms cancel and B is symmetric).

(iii) Equivalent extensions have cohomologous cocycles, and a coboundary $\delta\psi(x, y) = \psi(x) + \psi(y) + \psi(x + y)$ has zero diagonal ($\delta\psi(x, x) = \psi(0) = 0$), so the squaring form is an invariant of the class; (ii) gives surjectivity. Injectivity: if two extensions have equal squaring form Q , choose sections and cocycles φ, φ' ; both have diagonal Q , and both have symmetrization equal to the polar form of Q , so $d = \varphi + \varphi'$ is a symmetric cocycle with zero diagonal. The extension E_d is then abelian of exponent 2, i.e. an $\mathbb{F}_2$ -vector space, so $1 \rightarrow Z \rightarrow E_d \rightarrow V \rightarrow 1$ splits and d is a coboundary; the two extensions are equivalent.

(iv) Commutators generate $z^{\text{im } B}$, so E_Q is abelian iff $B = 0$. In general $Z(E_Q) = \pi^{-1}(\text{rad } B)$, since $\tilde{x}$ is central iff $B(x, \cdot) = 0$; thus $Z(E_Q) = Z$ iff B is nondegenerate. If so, $B \neq 0$ gives $[E, E] = Z$, and $\Phi(E) = E^2[E, E] = Z$ because all squares lie in Z and E/Z is elementary abelian. A nondegenerate alternating form exists only in even dimension. $\square$

Remark 10. This is the characteristic-2 Heisenberg picture: E_Q is the Heisenberg group of (V, B) and Q selects the central extension; cohomologically, (iii) is the standard identification of $H^2(V; \mathbb{F}_2)$ with quadratic maps $V \rightarrow \mathbb{F}_2$ [32]. The same finite-quadratic-module data drives Weil S/T matrices on discriminant forms (implemented in the integral layer of [1]). Note that the dictionary is exactly the characteristic-2 datum of Section 2.2 in group clothing: the diagonal q_i (squares) and the off-diagonal b_{ij} (commutators) are independent central data, and collapsing them is collapsing E_Q onto an abelian quotient.

Write H for the hyperbolic plane ($Q(e_1) = Q(e_2) = 0$, $B(e_1, e_2) = 1$; $\text{Arf} = 0$) and A for the anisotropic plane ($Q \equiv 1$ on $V \setminus \{0\}$; $\text{Arf} = 1$), and $\circ$ for the central product (identify the centers).

Lemma 2 (Arf = the two types). *Let B be nondegenerate, $\dim V = 2r \geq 2$.*

- (i) $E_{Q \perp Q'} \cong E_Q \circ E_{Q'}$.
- (ii) $E_H \cong D_8$ and $E_A \cong Q_8$.
- (iii) *Every extraspecial 2-group of order 2^{1+2r} is isomorphic to E_Q for a nondegenerate Q , and $E_Q \cong E_{Q'}$ iff $(V, Q) \cong (V', Q')$ iff the ranks and Arf invariants agree [13, 3]. Hence there are exactly two extraspecial groups of order 2^{1+2r} :*

$$\begin{aligned} E_{2^{1+2r}}^+ &= D_8^{\circ r} & (\text{Arf} = 0, Q \cong H^{\perp r}), \\ E_{2^{1+2r}}^- &= D_8^{\circ(r-1)} \circ Q_8 & (\text{Arf} = 1, Q \cong H^{\perp(r-1)} \perp A). \end{aligned}$$

In particular $D_8 \circ D_8 \cong Q_8 \circ Q_8$, the group avatar of $H \perp H \cong A \perp A$ (additivity of Arf).

- (iv) (Census.) With $\varepsilon = (-1)^{\text{Arf } Q}$,

$$\#\{g \in E_Q : g^2 = 1\} = 2 \#\{x : Q(x) = 0\} = 2^{2r} + \varepsilon 2^r,$$

so the element-order census of E_Q is the zero-count bias (2) read multiplicatively, and it distinguishes the two types.

Proof. (i) The cocycle $\varphi \oplus \varphi'$ on $V \oplus V'$ is bilinear with diagonal $Q \perp Q'$, and $(s, (x, x')) \mapsto [(s, x), (0, x')]$ is an isomorphism onto $(E_Q \times E_{Q'}) / \langle (z, z') \rangle$.

(ii) E_H has order 8, is nonabelian ($B \neq 0$), and has a noncentral involution (any lift of e_1 squares to $z^{Q(e_1)} = 1$), so $E_H \cong D_8$; in E_A every noncentral element squares to z (as $Q \equiv 1$ off 0), so z is the unique involution and $E_A \cong Q_8$.

(iii) For E extraspecial, $V = E/Z(E)$ is elementary abelian (as $\Phi(E) = Z(E)$), so Lemma 1 applies to $1 \rightarrow Z(E) \rightarrow E \rightarrow V \rightarrow 1$ and gives $E \cong E_{Q_E}$ with B_{Q_E} nondegenerate ($Z(E)$ exactly central). If $\psi: E_Q \rightarrow E_{Q'}$ is an isomorphism, it preserves the (characteristic) centers, fixes $z \mapsto z'$ (the unique nontrivial central element), and induces an $\mathbb{F}_2$ -linear map $\bar{\psi}: V \rightarrow V'$ with $Q'(\bar{\psi}x)$ read from $\psi(\tilde{x})^2 = \psi(\tilde{x}^2)$ – an isometry. Conversely an isometry g transports the cocycle: $\varphi' \circ (g \times g)$ has diagonal $Q' \circ g = Q$, hence by Lemma 1(iii) yields an extension equivalent to E_Q , and $(s, x) \mapsto (s, gx)$ closes the isomorphism. Dickson’s classification of nonsingular quadratic forms over $\mathbb{F}_2$ (two isometry classes per even rank, separated by Arf) finishes; the central-product normal forms follow from (i), (ii) and Witt decomposition.

(iv) The two lifts of x both square to $z^{Q(x)}$, so they are involutions or the identity iff $Q(x) = 0$; now apply the zero-count formula (2). The counts differ for $\varepsilon = \pm 1$, so $E^+ \not\cong E^-$. $\square$

Remark 11 (the integral shadow). The same extension data has an integral-lattice avatar, machine-verified through the discriminant-form pipeline of [1]: the Gold nonsingular core of rank $2r$ and Arf ε is the discriminant form of the even lattice $U(2)^r$ ($\varepsilon = 0$) or $U(2)^{r-1} \oplus D_4$ ($\varepsilon = 1$); the Milgram phase is $4 \cdot \text{Arf} \bmod 8$; the Weil S -matrix prefactor is $(-1)^{\text{Arf}}$; and the Frobenius–Schur indicator of the Heisenberg representation of E_Q is $(-1)^{\text{Arf}}$. The metaplectic relation $(ST)^3 = S^2$ holds iff the diagonal T uses a quadratic refinement of the correct Arf class – it selects the Arf *class*, not the member. So the integral bridge carries exactly one game-relevant bit; it cannot, by itself, distinguish the Gold form from any other refinement in its class.

9.3 The abelian obstruction

Lemma 3 (abelian obstruction). *Let M be a commutative monoid, $Z = \{1, z\} \subseteq M$ a two-element subgroup, $N \subseteq M$ a submonoid containing Z , and $\pi: N \twoheadrightarrow V$ a surjective monoid homomorphism onto an $\mathbb{F}_2$ -vector space with $\pi^{-1}(\pi(m)) = mZ$ for all $m \in N$. Then the squaring form $Q: V \rightarrow \mathbb{F}_2$, defined by $\tilde{m}^2 = z^{Q(x)}$ for any $\tilde{m} \in \pi^{-1}(x)$, is well defined and $\mathbb{F}_2$ -linear; its polar form vanishes identically. Consequently no commutative monoid realizes, through its own squaring map, a quadratic form with nonzero – a fortiori nondegenerate – characteristic-2 polar form. Equivalently, by Lemma 1(iv): if $B \neq 0$, the extension E_Q is nonabelian and admits no model (N, Z, π) inside any commutative monoid.*

Proof. $\pi(m^2) = 2\pi(m) = 0$, so $m^2 \in \pi^{-1}(0) = Z$, and $(zm)^2 = z^2m^2 = m^2$, so Q is well defined. Commutativity gives $(mn)^2 = m^2n^2$; since $\tilde{m}\tilde{n}$ is a preimage of $x+y$, this is $Q(x+y) = Q(x)+Q(y)$, i.e. $B \equiv 0$. $\square$

Theorem B (commutative squaring collapse). *In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular the extraspecial squaring map Q (for $B \neq 0$) cannot live in, map into, or pull back through any commutative game monoid. This covers misère quotients, turn-parity counters, and all coin-turning configuration sums.*

Proof. Immediate from Lemma 3: commutativity gives $(mn)^2 = m^2n^2$, so the squaring form is additive and $B \equiv 0$. $\square$

Corollary 5 (misère quotients: the kernel shadow is linear). *Let $\mathcal{Q}$ be a finite misère quotient – a finite commutative monoid – with kernel K , the maximal subgroup of $\mathcal{Q}$ (the mutual-divisibility class of the product of all idempotents) [31].*

- (i) *Every configuration (N, Z, π) as in Lemma 3 inside $\mathcal{Q}$ has linear squaring form: the intrinsic multiplication of a misère quotient cannot supply a quadratic refinement with $B \neq 0$, on K or anywhere else in $\mathcal{Q}$.*
- (ii) *Under the regularity hypothesis of [31, Thm. 6.4] (satisfied by the regular finite quotients arising in practice), $K \cong (\mathbb{Z}/2)^k$ is the group of normal-play Grundy values and the P -portion meets K in the XOR-linear normal-play set.*
- (iii) *On the smallest wild quotient $R_8 = \langle a, b, c \mid a^2 = 1, b^3 = b, bc = ab, c^2 = b^2 \rangle$ with $P = \{a, b^2\}$: the idempotents are $\{1, b^2\}$, the kernel is $K = \{b^2, b, ab, ab^2\} \cong (\mathbb{Z}/2)^2$ with identity b^2 , $P \cap K = \{b^2\} \mapsto \{0\}$ is linear, and the genuinely misère P -element a lies outside K . (Machine-verified: `misere_kernel` in [1].)*

So the linear obstruction observed on R_8 is forced by commutativity, not an accident of the example: a nondegenerate polar form is the commutator pairing of a nonabelian group, and a misère quotient has none to offer.

Proof. (i) is Lemma 3 applied to submonoids of $\mathcal{Q}$, which are commutative. (ii) is quoted from [31]. (iii) is a finite verification. $\square$

Theorem C (group misère quotients are tiny). *Assume [31, Thm. 6.4] with its regularity hypothesis. If the misère indistinguishability quotient $\mathcal{Q}(\mathcal{A})$ of a set of impartial games is a group, then $|\mathcal{Q}(\mathcal{A})| \leq 2$.*

Proof sketch. Doubling lemma: in a group quotient every element is cancellable, and any position of Grundy value 2 doubles to a misère- P position, contradicting Grundy-distinctness from 0; the quoted structure theorem then pins the quotient to at most $\{1, a\}$. $\square$

Remark 12 (scope). Theorem C is conditional on the Plambeck–Siegel structure theorem as cited. One consequence for machine searches: an elementary abelian quotient $(\mathbb{Z}/2)^k$ with $k \geq 2$ is empty a priori, so the empirical emptiness of an exhaustive octal sweep (292 octal codes, heap cutoff 4, observed quotient orders 2, 6, 10, 12, 14; `octal_hunt` in [1]) is forced by the theorem rather than a sampling accident.

9.4 The E -equivariance screen

Proposition 4 kills rules that are blind to everything but the symplectic structure (Tier 1), while per-position evaluator circuits for Q_a realize the quadric tautologically (Tier 3). The extraspecial dictionary suggests a criterion strictly between the two: the rule should see the extension E_Q – which carries the diagonal data q_i structurally, as squares – but only up to its automorphisms, so that no basis, field structure, or evaluation circuit can be smuggled in.

Definition 4 (uniform rules; the three tiers). Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$ with polar form B , and $E = E_Q$, $\pi: E \rightarrow V$, $\Sigma = \{x \in V : Q(x) = 0\}$.

- (a) A *uniform rule on a finite set X* is a single move relation $M \subseteq X \times X$, fixed once for all positions; for normal play we require the move digraph acyclic, and outcomes are computed by the usual retrograde recursion. (For loopy or misère readings, replace the outcome recursion; the equivariance constraint below applies verbatim, since outcome classes are invariants of digraph automorphisms.)
- (b) A uniform rule on E is *E -equivariant* if every $\alpha \in \text{Aut}(E)$ is an automorphism of the move digraph. Its *shadow* is $\pi(P) \subseteq V$, where P is its P -set; it *realizes* $T \subseteq V$ if $\pi(P) = T$.
- (c) *Tier 1* ($\text{Sp}(B)$ -blind): a uniform rule on V invariant under $\text{Sp}(B)$, as in Proposition 4. *Tier 3* (Q_a -evaluation): a family of games $\{\Gamma_x\}_{x \in V}$ whose structure varies with the designated input x (e.g. the trace-circuit evaluator); this is not a uniform rule. *Tier 2 screen*: a route to the Gold quadric is Tier-2 natural *only if* its positions and moves lift to an E -equivariant uniform rule on E_{Q_a} – with Q_a restricted to its nonsingular core when the form is degenerate, matching the classifier’s radical discipline – that realizes $\{Q_a = 0\}$.

Proposition 5 (the screen sits strictly between the solved tiers). *Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, and $E = E_Q$.*

- (i) *The image of $\text{Aut}(E) \rightarrow \text{GL}(V)$ is exactly the orthogonal group $\text{O}(Q)$, with kernel the central twists $g \mapsto gz^{\ell(\pi g)}$, $\ell \in V^*$, which equal $\text{Inn}(E) \cong V$ because B is nondegenerate; thus $\text{Out}(E) \cong \text{O}(Q)$ (cf. [38]). The $\text{Aut}(E)$ -orbits on E are*

$$\{1\}, \quad \{z\}, \quad \pi^{-1}(\Sigma \setminus \{0\}), \quad \pi^{-1}(V \setminus \Sigma).$$

- (ii) *Hence the P -set of any E -equivariant uniform rule is a union of these four orbits, and its shadow is one of the eight unions of $\{0\}$, $\Sigma \setminus \{0\}$, $V \setminus \Sigma$. The quadric Σ is among them: the Tier-1 exclusion does not apply.*

(iii) (Tier 1 $\subsetneq$ Tier 2.) Every $\mathrm{Sp}(B)$ -invariant uniform rule on V pulls back along π to an E -equivariant rule with P -set π^{-1} of the original, whose shadow is therefore one of $\emptyset$, $\{0\}$, $V \setminus \{0\}$, V – never the quadric (Proposition 4 in this language). The containment is strict: the squaring rule

$$g \rightarrow h \text{ legal} \iff g^2 = z \text{ and } h^2 = 1$$

(“move from any order-4 position to any position of order at most 2”) is E -equivariant and acyclic, and its P -set is the involution locus $\{g : g^2 = 1\} = \pi^{-1}(\Sigma)$, with shadow exactly Σ .

(iv) (Tier 2 $\subsetneq$ Tier 3.) Up to the isometry induced by any group isomorphism, the family of E -equivariant rules and their shadows depends only on $(r, \mathrm{Arf} Q)$ (Lemma 2(iii)). In particular the screen cannot separate Gold forms of equal core rank and Arf invariant, and an E -equivariant rule has no access to m , the exponent a , the field multiplication, the coordinate frame $q_i = Q_a(e_i)$, or any evaluation circuit. Conversely, arbitrary subsets of V – all realizable by ad hoc acyclic lookup games and by Tier-3 evaluators – are not shadows of E -equivariant rules once they leave the eight-set list of (ii).

Proof. (i) Automorphisms preserve $Z = Z(E)$ and fix z ($\mathrm{Aut}(\mathbb{Z}/2) = 1$), so the induced map preserves the squaring form: $Q(\bar{\alpha}x)$ is read from $\alpha(\tilde{x})^2 = \alpha(\tilde{x}^2) = z^{Q(x)}$; the image lies in $O(Q)$. Surjectivity: an isometry g transports the standard cocycle as in the proof of Lemma 2(iii), producing an automorphism of E_Q inducing g . The kernel consists of maps $g \mapsto g z^{\ell(\pi g)}$ with ℓ linear (multiplicativity forces additivity of ℓ); inner automorphisms are exactly the twists $\ell = B(\bar{g}, \cdot)$, and nondegeneracy makes $\bar{g} \mapsto B(\bar{g}, \cdot)$ onto V^* . Orbits: $O(Q)$ is transitive on $\Sigma \setminus \{0\}$ and on $V \setminus \Sigma$ by Witt’s extension theorem in its characteristic-2 quadratic-space form [36] (both sets are nonempty for $r \geq 2$); the two lifts of any $x \neq 0$ are conjugate under conjugation by $\tilde{g}$ with $B(\bar{g}, x) = 1$; and $1, z$ are fixed by everything.

(ii) Outcome recursion is invariant under digraph automorphisms. Thus P is $\mathrm{Aut}(E)$ -invariant; apply (i) and project.

(iii) The pullback $M = \{(g, h) : (\pi g, \pi h) \in R\}$ of an $\mathrm{Sp}(B)$ -invariant rule R is preserved by $\mathrm{Aut}(E)$, since the induced action on V lies in $O(Q) \subseteq \mathrm{Sp}(B)$; it is acyclic when R is, and an easy induction along the acyclic rank gives $P(M) = \pi^{-1}(P(R))$. By the transitivity of $\mathrm{Sp}(B)$ on $V \setminus \{0\}$ for $r \geq 2$, $P(R)$ is a union of $\{0\}$ and $V \setminus \{0\}$. For the squaring rule: automorphisms preserve element orders, so the rule is E -equivariant; positions with $g^2 = 1$ are terminal, hence P ; positions with $g^2 = z$ have a move (to 1), hence N . Its shadow Σ satisfies $1 < |\Sigma| = 2^{2r-1} + \varepsilon 2^{r-1} < 2^{2r} - 1$ for $r \geq 2$, so it is none of the four Tier-1 shadows.

(iv) If $\psi: E_Q \rightarrow E_{Q'}$ is an isomorphism (it exists iff ranks and Arf agree, Lemma 2(iii)), then $M \mapsto (\psi \times \psi)(M)$ is a bijection between E -equivariant rules carrying P -sets to P -sets and shadows to shadows through the induced isometry $\bar{\psi}$; in particular it carries rules realizing $\{Q = 0\}$ to rules realizing $\{Q' = 0\}$. The shadow constraint of (ii) excludes all but eight subsets, while Tier-3 constructions realize any subset; the containment is proper. $\square$

Remark 13 (what the screen does and does not settle). Equivariance is a screen, not a construction. Over the *abstract* group E the screen is satisfiable – Proposition 5(iii) exhibits the squaring rule, which is nothing but the extraspecial squaring map of Lemma 1 read as a one-move relation. That rule consults only the multiplication of E , so it is a Q -evaluator exactly insofar as E itself is fed in by hand. The reframing replaces the request *find a play rule that computes Q_a* by the structural question *exhibit a game-native model of the extension E_{Q_a}* – positions built from game values, multiplication from game constructions – after which the rule layer is canonical. Lemma 3 is the sharp constraint on that search: no commutative value-world, in particular no misère-quotient

kernel (Corollary 5), can host E once $B \neq 0$. The noncommutativity must enter from a structurally available asymmetry, and the one that normal, misère, and partizan play all possess – and that the symmetric polar form B discards – is the first-/second-player asymmetry of the move relation. The proposition is standard mathematics; treating E -equivariance as *the* naturality criterion is interpretation. The weighted-source rule resolves the P-set problem covariantly without first constructing the entire abstract extension as a game object. Theorem 9 now supplies that stronger model by an oriented trace cocycle; the weighted-source theorem remains the play-semantic construction with the sharp access contract. (For $r = 1$ the screen degenerates: on the anisotropic plane $\Sigma = \{0\}$ and $O(Q) = \text{Sp}(B)$, so the statement is vacuous there; the Gold targets of interest have $r \geq 2$ cores.) Theorem E below sharpens the screen further: read as *invariance* it admits no live middle at all, so the E -structure must enter *covariantly*, in the dynamics.

9.5 The game-native Gold–Heisenberg model

The preceding question has a basis-free answer hidden in the Gold monomial itself. The symmetric polar form is not the multiplication carrier; one must retain its oriented bilinear half.

Let $K = \mathbb{F}_{2^m}$ be represented by its nimber values, let $a \geq 0$, and allow a scale $\lambda \in K$. Put

$$\phi_{\lambda,a}(x, y) = \text{Tr}_{K/\mathbb{F}_2}(\lambda xy^{2^a}), \quad Q_{\lambda,a}(x) = \phi_{\lambda,a}(x, x). \quad (11)$$

The scale $\lambda = 1$ is the Gold form Q_a used throughout the paper.

Theorem 9 (game-native Gold–Heisenberg extension). *On*

$$\mathcal{E}_{\lambda,a} = \mathbb{F}_2 \times K$$

define

$$(s, x) \diamond (t, y) = (s + t + \phi_{\lambda,a}(x, y), x + y). \quad (12)$$

Then:

(i) $\mathcal{E}_{\lambda,a}$ is a group, and with $z = (1, 0)$ and $\pi(s, x) = x$,

$$1 \longrightarrow \langle z \rangle \longrightarrow \mathcal{E}_{\lambda,a} \xrightarrow{\pi} K^+ \longrightarrow 1$$

is a central extension whose fibers are the two $\langle z \rangle$ -cosets. Its inverse is $(s, x)^{-1} = (s + Q_{\lambda,a}(x), x)$.

(ii) *Squares and commutators are exactly*

$$(s, x)^2 = z^{Q_{\lambda,a}(x)}, \quad (13)$$

$$[(s, x), (t, y)] = z^{B_{\lambda,a}(x, y)}, \quad B_{\lambda,a}(x, y) = \phi_{\lambda,a}(x, y) + \phi_{\lambda,a}(y, x). \quad (14)$$

Consequently this is the extension $E_{Q_{\lambda,a}}$ of Lemma 1; the diagonal constants are the squares of the basis lifts, rather than supplied data.

(iii) *The product is one fixed game construction on two registered game values. Represent s by 0 or $*$ and x by the nimber game $*x$. Equation (12) becomes*

$$(C, X) \diamond (D, Y) = (C \oplus D \oplus \text{Tr}_{\text{game}}((\ast\lambda) \otimes X \otimes \text{Frob}^a(Y)), X \oplus Y), \quad (15)$$

where $\ast\lambda = \ast 1$ in the unscaled case. Thus it uses only disjunctive sum, Turning Corners, diagonal Frobenius, and their trace iterate. It contains no q_i table, unary $Q_{\lambda,a}$ oracle, outcome test, or input-dependent rule.

(iv) The multiplication acts faithfully by game-built contexts. If $\mathcal{G}$ is the additive group of short partizan game values and $\mathcal{C} = \mathcal{G}^K$, set

$$(zF)(u) = -F(u), \quad (T_x F)(u) = (-1)^{\phi_{\lambda,a}(u,x)} F(u+x). \quad (16)$$

Then $T_x T_y = z^{\phi_{\lambda,a}(x,y)} T_{x+y}$, and $(s, x) \mapsto z^s T_x$ embeds $\mathcal{E}_{\lambda,a}$ into $\text{Aut}(\mathcal{C})$.

(v) The center is

$$Z(\mathcal{E}_{\lambda,a}) = \{(s, r) : s \in \mathbb{F}_2, r \in \text{rad } B_{\lambda,a}\}.$$

Hence the group is extraspecial exactly when the polar form is nondegenerate of positive rank. More generally, every positive-rank nonsingular complement W of the radical gives an extraspecial subgroup $\mathbb{F}_2 \times W$ with squaring form $Q_{\lambda,a}|_W$ and order $2^{1+\text{rank } B_{\lambda,a}}$.

Proof. Frobenius is additive and trace is linear, so $\phi = \phi_{\lambda,a}$ is biadditive. Therefore

$$\phi(x, y) + \phi(x + y, w) = \phi(y, w) + \phi(x, y + w),$$

the normalized cocycle identity. Expanding both bracketings of (12) proves associativity. The identity and displayed inverse follow directly (the additive groups have exponent two), and $\langle z \rangle$ is central with quotient K^+ and two-point fibers.

Putting the same input in both slots gives

$$(s, x)^2 = (\phi(x, x), 0) = (Q_{\lambda,a}(x), 0).$$

Swapping two factors in (12) changes only the central coordinate, by $\phi(x, y) + \phi(y, x)$. This proves (13)–(14) and identifies the extension by Lemma 1.

Equation (15) is simply the same formula with each field operation written in the game realization of Section 2.1; in particular the trace value lands in the central nimber pair $\{0, *\}$. For the context model, biadditivity gives

$$\begin{aligned} (T_x T_y F)(u) &= (-1)^{\phi(u,x) + \phi(u+x,y)} F(u+x+y) \\ &= (-1)^{\phi(u,x+y) + \phi(x,y)} F(u+x+y), \end{aligned}$$

which is the claimed multiplication. Let $\delta_0 \in \mathcal{C}$ take the integer-game value 1 at 0 and 0 elsewhere. The support of $z^s T_x \delta_0$ is the singleton $\{x\}$, and its nonzero value has sign $(-1)^{s+Q_{\lambda,a}(x)}$; hence the action is faithful.

Finally, (s, x) commutes with every (t, y) exactly when $B_{\lambda,a}(x, y) = 0$ for every y , proving the center formula. On a nonsingular complement the center is $\langle z \rangle$, a nonzero commutator generates that center, and $\Phi(G) = G^2[G, G] = \langle z \rangle$; this is precisely extraspeciality. $\square$

The theorem also corrects a harmless but real naming defect in the earlier question. The full unscaled Gold group is generally an *extraspecial-type extension*, not itself extraspecial, because the unscaled Gold polar form is singular. In the power-of-two nimber tower the repair is canonical rather than a choice of Witt complement.

Proposition 6 (canonical Gold radical quotient). *Let $\lambda = 1$, put $d = \gcd(2a, m)$, and let $R = \mathbb{F}_{2^d} \subseteq K$. Then*

$$\text{rad } B_a = R, \quad \text{rank } B_a = m - d.$$

If m is a power of two and $d < m$, then $Q_a|_R = 0$ and $\phi_a|_{R \times R} = 0$. Thus $H = \{(0, r) : r \in R\}$ is central and elementary abelian, and

$$\mathcal{E}_a/H \longrightarrow K/R$$

is a canonical extraspecial extension of order 2^{1+m-d} whose squaring form is the induced nonsingular Gold core. For $(m, a) = (4, 1)$ this quotient has order eight and, by the Arf-1 classification of the core, is Q_8 .

Proof. Trace adjointness and Frobenius invariance give

$$B_a(x, y) = \text{Tr}\left((x^{2^{m-a}} + x^{2^a})y\right),$$

with exponents read modulo m . The trace pairing is nondegenerate, so x is radical exactly when $x^{2^{2a}} = x$; the fixed field is $\mathbb{F}_{2^{\gcd(2a, m)}}$. This proves the rank formula.

When m is a power of two and $d < m$, the relative degree m/d is even. For $r, s \in R$, both rs^{2^a} and r^{1+2^a} lie in R , and absolute trace from K restricts to (m/d) times trace from R , hence vanishes. This proves both restrictions. Since R is the polar radical, H is central; the vanishing of ϕ on $R \times R$ makes it a subgroup, and the quotient quadratic form is well defined. Its polar form on K/R is nondegenerate, so Theorem 9(v) makes the quotient extraspecial. The last identification follows from its order and the already established Arf-1 core row. $\square$

Remark 14 (exact naturality boundary). This is a per-field, Frobenius-covariant structural model: $\phi_a(x^{2^j}, y^{2^j}) = \phi_a(x, y)$. It cannot be ambient-coherent through even-degree subfield inclusions, because absolute trace restricts to zero there; that is consistent with the global divisibility no-go of Appendix A. Nor can any section of $\mathcal{E}_a \rightarrow K$ be fixed by all extension automorphisms, since central twists exchange its two lifts. Neither stronger condition belonged to the question. The faithful context action of Theorem 9(iv) likewise does not collide with the E -translation no-go (Theorem H below): it represents the structural multiplication, not a play rule with moves $g \rightarrow gn$ for a fixed move set; once the extension is built, the play layer is the automorphism-equivariant squaring rule of Proposition 5(iii). A stronger demand – that the carrier be one untagged Conway value under a single pre-existing binary compound – would be a different problem: ordinal sum does not descend to game values, while unrestricted quotients of literal game trees can import any finite presentation and are therefore too permissive. Finally, “without a Q_a evaluator” cannot coherently forbid multiplication from revealing Q_a on a square – every model of E_{Q_a} does exactly that. The meaningful and met condition is that multiplication not import Q_a or its diagonal as an oracle.

9.6 Symmetry budgets

For the remaining results, fix a nondegenerate quadratic form Q on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, with polar form B and zero set $\Sigma = \{Q = 0\}$, and write $\text{AO}(Q) = \{x \mapsto gx + c : g \in \text{Sp}(B), Q(gx + c) = Q(x) \text{ for all } x\}$ for the affine stabilizer appearing below.

Theorem D (affine symmetry budget). $\text{Stab}_{\text{AGL}(V)}(\Sigma) = \text{AO}(Q)$; it contains no nontrivial pure translation (for a nonsingular core, $c \in \text{rad}(B) = 0$), and its pure-linear part is exactly $\text{O}(Q)$. Consequently every uniform rule whose move digraph admits even one affine automorphism outside $\text{AO}(Q)$ has P -set $\neq \Sigma$.

Proof. A Boolean-valued quadratic form is determined as a function by its zero set. Thus an affine map stabilizes Σ exactly when it preserves Q , which after polarization forces its linear part to preserve B . This proves $\text{Stab}_{\text{AGL}(V)}(\Sigma) = \text{AO}(Q)$. If the pure translation $x \mapsto x + c$ preserves Q , evaluation at $x = 0$ gives $Q(c) = 0$, and then

$$0 = Q(x + c) + Q(x) = B(x, c) \quad \text{for every } x.$$

Hence $c \in \text{rad}(B) = 0$. With $c = 0$, preserving Q is precisely membership in $\text{O}(Q)$. $\square$

Theorem E (the symmetry dial has no middle). *The classical Dye–Dieudonné maximality theorem gives that $O(Q)$ is maximal in $\mathrm{Sp}(B)$ [16, 14], and the $O(Q)$ -orbitals on $V \times V$ are exactly the Gram classes $(Q(u), Q(w), B(u, w))$ together with the degeneracy flags (Witt extension; see [17, §8]). Hence a rule invariant under any group strictly larger than $O(Q)$ is dead by Theorem D (its symmetry reaches outside $\mathrm{AO}(Q)$), while a rule fully $O(Q)$ - or $\mathrm{Aut}(E)$ -equivariant factors extensionally through pointwise Q - and B -evaluation. There is no invariance class strictly between dead (too coarse) and Gram-factoring (an evaluator): E -naturality must be covariance, not invariance.*

Corroboration. At $m = 4$, every one of the 648 (resp. 600) elements of $\mathrm{Sp}(4, 2)$ outside $O^+(4, 2)$ (resp. $O^-(4, 2)$) generates $\mathrm{Sp}(4, 2)$ together with the orthogonal group; orbitals enumerated exhaustively at $m = 4$ (`ao_orbitals` in [1]).

9.7 Access budgets

Theorem F (B -oracle lower bound). *Access model: legality of a move is decided from XOR combinations of the two positions and t fixed constant vectors, with oracle access to B only. If $t \leq 2r - 3$, then for every quadratic refinement Q' of B simultaneously there is a nonzero Q' -singular vector in the orthogonal complement of the constant span; the transvection it induces is a digraph automorphism lying outside $O(Q')$, and Theorem D kills the P -set. The bound is tight: at $t = 2r - 2$ with anisotropic complement the argument fails. Hence the Gold diagonal framing $q_i = Q_a(e_i)$, which supplies $2r$ frame constants, is within two dimensions of information-theoretically forced.*

Proof. Let W be the span of the t constants. Since $\dim W^\perp \geq 3$, the restriction of Q' to $W^\perp$ has a nonzero singular vector v (a totally anisotropic binary quadratic space has dimension at most two). The symplectic transvection $T_v x = x + B(x, v)v$ fixes every constant. Writing $b = B(x, v)$ and using $Q'(v) = 0$ gives

$$Q'(T_v x) = Q'(x) + bQ'(v) + bB(x, v) = Q'(x) + B(x, v).$$

Thus T_v preserves B and every allowed oracle expression but does not preserve Q' ; nonsingularity of B supplies an x with $B(x, v) = 1$. Theorem D applies. When $\dim W^\perp = 2$, its restriction may be anisotropic, explaining the stated sharp boundary of this argument. $\square$

Corroboration. The exact escape boundary $t = 2r - 2$ was machine-checked at $m = 4$; an end-to-end spot check retrograde-solved 50 random $t = 1$ B -oracle rules (`skeptic_nogo_check`, `nogo_verify` in [1]).

Theorem G (B -local flip rules). *Any rule of the form “flip d at v iff $f(d, B(v, d))$ ” is automatically undirected, because B is alternating: $f(d, B(v + d, d)) = f(d, B(v, d) + B(d, d)) = f(d, B(v, d))$. For undirected loopy games the outcome partition collapses: $\mathrm{Win} = \emptyset$, $\mathrm{Loss} =$ the isolated vertices = an affine flat, and $\mathrm{Loss} \cup \mathrm{Draw} \neq \Sigma$ for $r \geq 2$ (the count $|\Sigma| = 2^{r-1}(2^r + \varepsilon)$ has an odd factor > 1). This covers the Loss , Draw , and $\mathrm{Loss} \cup \mathrm{Draw}$ targets at once. The symmetric- B rule probed in [1] (`loopy_quadric`) is the special case $f(d, \beta) = \beta$; its $(4, 1)$ “hit” is the radical coincidence $|\{Q_1 = 0\}| = 4 = |R(B)|$ at the vacuous instance of Remark 4.*

Verification. Undirectedness is the two-line identity above; the loopy outcome collapse including the Draw -set gap is machine-checked at $m = 4$ (`skeptic_nogo_check` in [1]).

Theorem H (E -translation no-go). *For $r \geq 2$, no left- E -equivariant rule on $E = E_Q$ with moves $g \rightarrow gn$, $n \in N$ for a fixed nonempty $N \subseteq E$, has P -set equal to the involution locus $I = \{g : g^2 = 1\} = \pi^{-1}(\Sigma)$.*

Proof. $I \cdot I = E$: for every $v \neq 0$, the count $\#\{x : Q(x) = 0 = Q(x + v)\} = 2^{2r-2} + (-1)^{\text{Arf } Q} 2^{r-1}$ is positive for $r \geq 2$ (character sum), so every $n \in N$ factors as $n = gh$ with $g, h \in I$. Then $g \in I$, and $g \cdot n = g(gh) = g^2h = h \in I$: the move $g \rightarrow gn$ is an edge from I to I . A normal-play P -set admits no P -to- P edge, so $P \neq I$. $\square$

Verification. Cross-checked on the bent $(8, 1, \lambda = 2)$ extraspecial group (`extraspecial_core` in [1]).

9.8 Normal-play rigidity and the synthesis

The strongest constraint is not any single architecture kill but a rigidity statement inside a declared access class. For this subsection, call a rule *in-class* only if it is a *single-board vector-position rule*: its positions are exactly the vectors of V . In addition, the quadratic refinement is quarantined behind a weight-bounded oracle (the (w_0, c) metering of Definition 1), and the rule is *refinement-uniform*: it realizes the target for every refinement q of B in the declared family, not just the Gold one.

Theorem 10 (normal-play rigidity, in-class). *Under the bounded-framing oracle model with refinement uniformity, every legal edge between vector positions of Hamming weight $> cw_0$ must flip Q – in normal, misère, and loopy-Loss semantics alike. Consequently every in-class realizer of Σ in those semantics is a Q -alternating ender: play telescopes Q , and the outcome is the parity of $Q(x)$.*

Proof sketch. The case $Q(v) = Q(w) = 0$ of a non-flipping edge is forbidden directly by the outcome structure in all three readings. The case $Q(v) = Q(w) = 1$ is reduced to it by an adversary substitution: replace q by $q' = q + \ell$ with ℓ a linear form annihilating every queried functional but not the dense indicators; refinement uniformity transports the rule to q' , where the edge becomes a forbidden 0-to-0 edge. $\square$

Theorem 11 (no live middle, relative to the quarantine). *Let Q have nonsingular core of rank $2r \geq 4$. No fixed uniform single-board vector-position rule satisfying*

- (S) *no affine digraph automorphism outside $\text{AO}(Q)$,*
- (A) *legality from B -oracle queries at $t \leq 2r - 3$ frame constants,*
- (C) *no commutative squaring route for the provenance of Q ,*
- (E) *no left- E -equivariant structure on the extraspecial group,*
- (R) *bounded-framing oracle access with refinement uniformity,*

has normal-play P , misère P , or loopy Loss target equal to Σ with any bulk strategic content: under (R), every legal edge between dense positions flips Q , and all non-alternating strategic content is confined to a Hamming ball of radius cw_0 .

Proof. The conclusion is Theorem 10 applied under (R): in each covered semantics, every legal edge between positions of Hamming weight $> cw_0$ flips Q , so any realizer in the class is a Q -alternating ender, with all non-alternating strategic content confined to the stated Hamming ball. The remaining hypotheses pin the class against the preceding catalogue: (S), (C), and (E) exclude the symmetry, provenance, and translation routes of Theorems D, B, and H, while (A) identifies the bounded B -oracle public-access regime that Theorem F excludes when no refinement queries are available. $\square$

Remark 15 (scope of the no-go theorem). Theorem 11 is relative to its hypotheses. It does not cover:

1. *Loopy-Draw semantics*: the substitution contradiction of Theorem 10 does not fire (Draw-to-Draw edges are legal).
2. $t \geq 2r - 2$ *with anisotropic complement*: escapes Theorem F at the symmetry level.
3. *Frobenius-aware access*: the Galois group lies inside $O(Q_a)$, so all symmetry methods are silent; Theorem F’s model explicitly excludes Frobenius-twisted queries.
4. *State-extended rules*: the weighted-source construction has FIFO positions $(U, \mathcal{Q}, \text{ko}, \sigma)$ rather than positions in V . This is its escape from the single-board hypothesis; it still satisfies N1 uniformly over every refinement, so game-built Gold sourcing is not an escape from torsor uniformity.
5. *Rank $r = 1$ and radical-anisotropic degenerate layers*: the character-sum count in Theorem H vanishes, and small extraspecial groups can be realized.

The main construction uses the fourth route: a public Witt-frame lift with a genuinely larger FIFO state space. Sections 4 and 6 replace vocabulary quarantine by the invariant transcript-span boundary, and Section 5 attains that boundary with an explicit local rule. The list therefore records the theorem’s scope, not unresolved premises of the realization.

10 The formal boundary

The headline realization and observation bound separate into six independently checked components; a seventh module checks the auxiliary padding obstruction:

- `formal/Ogdoad/GoldDiagonal.lean`: trace blocks and the Artin–Schreier identities behind the all-exponent source;
- `formal/Ogdoad/FifoMatching.lean`: the original both-seat matching strategy;
- `formal/Ogdoad/ImpartialRealizer.lean`: the pass-free transition, exact even clock, both-seat safe-front induction, and impartial tail;
- `formal/Ogdoad/GoldMatchingAlgebra.lean`: the adapted-basis expansion;
- `formal/Ogdoad/GoldNoEvaluator.lean`: the observation lower bound;
- `formal/Ogdoad/GoldBlockCompression.lean`: the induced-instance algebra behind the every-width attainment;
- `formal/Ogdoad/GoldForkPadding.lean`: outcome-preserving padding.

Three further modules support the closing sections: `formal/Ogdoad/BrownGame.lean` kernel-checks the algebraic core of Section 8 – the Brown split with its converse, the two-divisible global collapse, and the selector outcome table with its decoder, each arena abstracted through its outcome-bit contract – and `formal/Ogdoad/GameExterior.lean` kernel-checks the root-collapse and coefficient-faithfulness core of Appendix A; and `formal/Ogdoad/GoldExtraspecial.lean` kernel-checks the biadditive cocycle group, square and commutator laws, center criterion, and order-four lifts of Theorem 9. All modules build against `mathlib` and live in the companion repository [1].

The paper supplies the concrete Witt basis, the original-frame coefficient transport, and the synthesis of those components. There is no single end-to-end Lean theorem constructing the complete arena. In particular, the pass-free Lean theorem begins from a supplied binary score, is pointwise

in the scoring submatching, and does not reify the one public policy across all submatchings or encode weighted source OPEN charges. The displayed safe-front proof fixes that policy on the full public potential matching and proves every CLOSE clean; the adapted-basis identity then supplies the paper-level weighted-source composition. Similarly, `formal/Ogdoad/GoldExtraspecial.lean` is universal in an abstract biadditive ϕ ; the concrete nimber-field trace specialization of Theorem 9 is the paper-level step, independently exercised by the repository’s tested arithmetic surface. These are the exact remaining formalization boundaries, not open mathematical premises.

The arbitrary-graph isolated-dummy FIFO theorem remains open and is studied in the companion note cited in Section 5. It would strengthen FIFO combinatorics but does not strengthen Theorem 2, whose boards are matchings plus isolates. The all-exponent diagonal source is supplied by Theorem 6; it is not an additional hypothesis.

11 Conclusion

Every quadratic refinement of every finite alternating binary space therefore has a uniform bounded-access *impartial* normal-play realization. A public Witt frame makes the polar graph a matching, weighted source pairs preserve original-frame singleton locality, the safe-front theorem forces the exact charge in every dimension, the pass-free core has fixed even tempo, and one impartial charge tail compiles that bit to a P/N outcome. For the motivating Gold trace forms, nimber operations source the diagonal and the Arf invariant with radical data gives the resulting zero-set bias. The oriented trace half $\phi_a(x, y) = \text{Tr}(xy^{2^a})$ simultaneously gives a structural multiplication on two nimber-game registers: squares recover Q_a , commutators recover B_a , and the canonical power-of-two radical quotient is extraspecial.

The surrounding negative results make the access boundary sharp. Linear coin-turning reaches the lexicode floor but not a genuine quadratic zero set; the extraspecial extension identifies the missing noncommutative datum; symmetric, commutative, and bounded-total-observation architectures cannot carry it uniformly; and extensional fork tests cannot define naturality. Weighted-source Witt–FIFO meets the necessary observation span with singleton support, and its block aggregation (Corollary 2) closes the boundary at every fixed width.

The arbitrary-graph isolated-dummy FIFO linking conjecture – a strict strengthening of Theorem 1, no longer load-bearing here – remains a separate combinatorial problem. The game-native extension question is closed by Theorem 9: the covariant play realization and the structural group model are now distinct proved faces of the same Gold quadratic datum.

References

- [1] a9lim, *ogdoad: Clifford algebras over game-adjacent scalar worlds*. Rust library with experiment scripts and Lean 4 formal supplement, 2026. <https://github.com/a9lim/ogdoad>.
- [2] H. Altman, P. Lipparini, *A ring structure on the class of combinatorial games*, arXiv:2604.27847 (2026).
- [3] C. Arf, *Untersuchungen über quadratische Formen in Körpern der Charakteristik 2, I*, J. Reine Angew. Math. **183** (1941), 148–167.
- [4] E. R. Berlekamp, J. H. Conway, R. K. Guy, *Winning Ways for Your Mathematical Plays*, 2nd ed., A K Peters, 2001–2004.
- [5] A. Bouchet, *Isotropic systems*, European J. Combin. **8** (1987), 231–244.
- [6] M. Brandenburg, *Algebraic games–playing with groups and rings*, Internat. J. Game Theory **47** (2018), 417–450.

- [7] R. Brijder, L. Traldi, *Isotropic matroids II: Circle graphs*, Electron. J. Combin. **23** (2016), Paper 4.2; arXiv:1504.04299.
- [8] E. H. Brown, Jr., *Generalizations of the Kervaire invariant*, Ann. of Math. **95** (1972), 368–383.
- [9] C. Carlet, *Boolean Functions for Cryptography and Coding Theory*, Cambridge University Press, 2021.
- [10] M. Cohn, A. Lempel, *Cycle decomposition by disjoint transpositions*, J. Combin. Theory Ser. A **13** (1972), 83–89.
- [11] J. H. Conway, *On Numbers and Games*, Academic Press, 1976.
- [12] J. H. Conway, N. J. A. Sloane, *Lexicographic codes: error-correcting codes from game theory*, IEEE Trans. Inform. Theory **32** (1986), 337–348.
- [13] L. E. Dickson, *Linear Groups, with an Exposition of the Galois Field Theory*, Teubner, 1901.
- [14] J. Dieudonné, *La géométrie des groupes classiques*, 3rd ed., Ergebnisse der Mathematik **5**, Springer, 1971.
- [15] E. Duchêne, A. Parreau, M. Rigo, *Deciding game invariance*, arXiv:1408.5274 (2014).
- [16] R. H. Dye, *On the maximality of the orthogonal groups in the symplectic groups in characteristic two*, Math. Z. **172** (1980), 203–212.
- [17] R. Elman, N. Karpenko, A. Merkurjev, *The Algebraic and Geometric Theory of Quadratic Forms*, AMS Colloquium Publications **56**, American Mathematical Society, 2008.
- [18] T. S. Ferguson, *Game Theory*, Part I, Ch. 5: Coin Turning Games, UCLA online lecture notes.
- [19] A. S. Fraenkel, *Error-correcting codes derived from combinatorial games*, in *Games of No Chance*, MSRI Publ. **29**, Cambridge University Press, 1996, 417–431; arXiv:math/9504211.
- [20] L. Fuchs, *Abelian Groups*, Springer Monographs in Mathematics, Springer, 2015.
- [21] R. Gold, *Maximal recursive sequences with 3-valued recursive cross-correlation functions*, IEEE Trans. Inform. Theory **14** (1968), 154–156.
- [22] D. Gorenstein, *Finite Groups*, 2nd ed., Chelsea Publishing, New York, 1980.
- [23] T. Hellese, P. V. Kumar, *Sequences with low correlation*, in *Handbook of Coding Theory*, Vol. II, Elsevier, 1998, 1765–1853.
- [24] W. Johnson, *The combinatorial game theory of well-tempered scoring games*, arXiv:1112.3610 (2011).
- [25] W. Johnson, *Combinatorial game theory, well-tempered scoring games, and a knot game*, arXiv:1107.5092 (2011).
- [26] I. Kaplansky, *Infinite Abelian Groups*, University of Michigan Press, 1954.
- [27] H. W. Lenstra, Jr., *Nim multiplication*, Séminaire de Théorie des Nombres de Bordeaux, 1977–1978.
- [28] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.
- [29] D. Moews, *The abstract structure of the group of games*, in *More Games of No Chance*, MSRI Publ. **42**, Cambridge University Press, 2002, 49–57.
- [30] V. Ovsienko, *Real Clifford algebras and quadratic forms over $\mathbb{F}_2$: two old problems become one*, The Mathematical Intelligencer **38** (2016); arXiv:1601.07664.
- [31] T. E. Plambeck, A. N. Siegel, *Misere quotients for impartial games*, J. Combin. Theory Ser. A **115** (2008), 593–622.
- [32] D. Quillen, *The mod 2 cohomology rings of extra-special 2-groups and the spinor groups*, Math. Ann. **194** (1971), 197–212.

- [33] K.-U. Schmidt, $\mathbb{Z}_4$ -valued quadratic forms and quaternary sequence families, IEEE Trans. Inform. Theory **55** (2009), 5803–5810.
- [34] L. Sok, M. Shi, P. Solé, *Decomposition of bent generalized Boolean functions*, arXiv:1611.06357 (2016).
- [35] P. Solé, N. Tokareva, *Connections between quaternary and binary bent functions*, Cryptology ePrint Archive, Report 2009/544 (2009).
- [36] D. E. Taylor, *The Geometry of the Classical Groups*, Sigma Series in Pure Mathematics **9**, Heldermann Verlag, 1992.
- [37] C. T. C. Wall, *Quadratic forms on finite groups, and related topics*, Topology **2** (1963), 281–298.
- [38] D. L. Winter, *The automorphism group of an extraspecial p -group*, Rocky Mountain J. Math. **2** (1972), 159–168.
- [39] J. A. Wood, *Witt’s extension theorem for mod four valued quadratic forms*, Trans. Amer. Math. Soc. **336** (1993), 445–461.
- [40] Z. Zhang, Y. Shi, *On the parity complexity measures of Boolean functions*, arXiv:1004.0436 (2010).

A The game-exterior obstruction

Altman and Lipparini ask whether Clifford algebras can be constructed game-theoretically [2, Problem 5.3(j)]. Their Remark 3.5 also isolates the basic boundary: Conway equivalence makes games an abelian group under disjunctive sum, but does not make Conway product well-defined on arbitrary game values. The theorem below gives a negative answer to that question for a precise and natural Clifford interpretation: an additive grade-one realization with coefficient-faithful square and polar relations that is coherent under adjoining short-game roots. It does not rule out a different multiplication or a different naturality contract.

Write ShUg for the additive group of short partizan games. An ambient-coherent datum must mean either one datum on all of ShUg, or a compatible system over *every* finitely generated subgroup (equivalently, over a cofinal directed subsystem). In the latter formulation, inclusion maps must inject the coefficient rings and algebras and preserve the grade-one maps, squares, and anticommutators. The filtered colimits of those rings, algebras, and groups glue such a system to the former global datum, and restriction gives the converse. This quantifier is essential: a family confined to selected subgroups need not see an actual root. In particular, the nonzero Gold forms on finite nimber subgroups would otherwise be counterexamples to the theorem.

Let R be a commutative ring, C an R -algebra, and $\iota: R \hookrightarrow C$ an injective coefficient map. A coefficient-faithful datum consists of an additive map $i: \text{ShUg} \rightarrow C$ and functions Q, B satisfying

$$i(x)^2 = \iota(Q(x)), \quad i(x)i(y) + i(y)i(x) = \iota(B(x, y)).$$

The equations and additivity already force the usual polarization and biadditivity identities. In particular, the diagonal anticommutator gives

$$\iota(B(y, y)) = 2i(y)^2 = \iota(2Q(y)), \quad B(y, y) = 2Q(y), \tag{17}$$

and hence $Q(2y) = 4Q(y)$.

Theorem 12 (ambient-coherent game-exterior obstruction). *In every ambient-coherent coefficient-faithful Clifford datum on ShUg,*

$$Q(x), B(x, y) \in \bigcap_{k \geq 0} 4^k R \quad (x, y \in \text{ShUg}).$$

Moreover, for every torsion game t and every short game x ,

$$Q(t) = 0, \quad B(t, x) = B(x, t) = 0.$$

Consequently the quadratic datum factors through the torsion-free quotient.

Proof. Moews's proof of his structure theorem uses Norton halving, and gives 2-divisibility of ShUg ; all finite orders are powers of 2 by Conway's Theorem 92 [11, Theorem 92], as quoted by Moews [29]. Thus, for every k , write $x = 2^k u$ and $y = 2^k v$. Iterating Equation (17) gives

$$Q(x) = 4^k Q(u), \quad B(x, y) = 4^k B(u, v),$$

where the second equality uses the forced biadditivity of B . This proves the intersection assertion.

For the torsion assertion, let $nt = 0$, with n a power of 2. Choose y with $ny = t$. Since $ni(t) = i(nt) = 0$,

$$i(t)^2 = i(t)i(ny) = (ni(t))i(y) = 0,$$

so coefficient faithfulness gives $Q(t) = 0$. To kill the polar value, choose z with $nz = x$; no root of t is needed here:

$$\begin{aligned} i(t)i(x) + i(x)i(t) &= i(t)i(nz) + i(nz)i(t) \\ &= (ni(t))i(z) + i(z)(ni(t)) = 0. \end{aligned}$$

Injectivity gives $B(t, x) = 0$, and symmetry gives $B(x, t) = 0$. □

The diagonal identity in (17) is where coefficient faithfulness is indispensable: it equates two elements of C , then uses injectivity of ι to recover an equality in R . The torsion anticommutator argument itself uses only additivity of i and bilinearity of multiplication.

Corollary 6 (coefficient consequences). *Over $\mathbb{Z}$, in characteristic 2, or over $\mathbb{Z}/4$, the quadratic and polar data are identically zero. In every coefficient ring, all torsion generators are exterior/nilpotent and polar-orthogonal to every grade-one generator.*

Proof. In each of the three named targets, $\bigcap_{k \geq 0} 4^k R = 0$. The remaining statement is the torsion part of Theorem 12. □

There is a concrete root behind the abstract input. In Moews's notation, if N_H is the least nonnegative integer with $H + 2N_H > 0$, Norton multiplication gives

$$\phi(H) = \frac{1}{2} \cdot (H + 2N_H) - N_H, \quad 2\phi(H) = H.$$

For $H = *$, $N_* = 1$, so

$$\phi(*) = \frac{1}{2} \cdot (* + 2) - 1, \quad 2\phi(*) = (* + 2) - 2 = *.$$

Indeed, the defining Norton rule is additive in its real multiplier, so $2(\frac{1}{2} \cdot U) = 1 \cdot U = U$; this verifies the displayed halving identity directly. This is the step used inside Moews's proof of Theorem 9, and is independently recorded in [25, Corollary 6.2.5].

What remains. Over $\mathbb{Z}$ even the torsion-free dyadic directions admit no nonzero additive scalar readout, since $\text{Hom}(\mathbb{Z}[1/2], \mathbb{Z}) = 0$. The mean-value homomorphism does survive after passing to a target containing the relevant dyadic values (in particular, one in which 2 is invertible); its square and polarization give the usual free-direction examples. Nor is the obstruction a claim about every quotient: for example,

$$\mathbb{Z}[1/2]/3\mathbb{Z}[1/2] \cong \mathbb{Z}/3$$

carries ordinary quadratic data over $\mathbb{F}_3$. These elementary divisible/torsion distinctions belong to the standard abelian-group calculus [20, 26]; Wall’s quadratic and linking forms on finite groups are the classical contrasting setting [37].

Three escapes are therefore real but deliberately outside the theorem. First, a hand-supplied table on a root-incomplete subgroup is the existing engineering API of [1]. Second, the Gold forms in the nimber core are game-native for a different, operation-based naturality condition: they use that core’s addition, nim-product, Frobenius, and trace. Requiring extension across all short-game inclusions is one naturality condition, not the weakest possible one; it is chosen here because it asks whether a value-level Clifford datum survives adjoining the actual partizan roots of its generators. Under that condition those Gold forms cannot extend. Finally, a construction without an additive grade-one map is not constrained by this theorem, but is no longer the Clifford datum just defined.

The algebraic root-collapse and coefficient-faithfulness core is kernel-checked in `formal/Ogdoad/GameExterior.lean`. The passage from a cofinal family to its colimit, and the sharpened $\bigcap_k 4^k R$ conclusion, are paper-level arguments here.